Refractory High-Entropy Alloys and Hybrid Additive Routes for Metallurgy in Lunar Mare, Highlands and KREEP Terranes

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Abstract

Sustained lunar operations will depend on turning actual lunar materials, mare basalts, feldspathic highlands and KREEP-rich crust,into usable metals and components. These reservoirs offer distinct chemistries: FeO- and TiO_2-rich mare basalts with abundant ilmenite ($FeTiO_3$) as a source of Fe, Ti and O; plagioclase-dominated highlands rich in Al_2O_3 and CaO; and the Procellarum KREEP Terrane (PKT) and related KREEP-bearing units enriched in K, rare-earth elements, P and heat-producing U/Th. Within ESA's FIRST! programme (SPARK – Strong Performance Alloys for Rocket Kinetics), Bimo Tech and ArianeGroup are developing RHEAs and related systems for oxygen-rich preburners and other high-temperature propulsion components. These alloys are produced on Earth via vacuum melting, powder metallurgy and hybrid additive/subtractive routes (SLM/DED plus near-net-shape machining), and tested up to $\sim$1 200 °C for strength retention, oxidation resistance and microstructural stability in engine-relevant environments. This contribution shows how such Earth-qualified RHEAs and processes form a direct stepping stone toward lunar metallurgy architectures that respect real resource maps. Process windows and feedstock specifications that remain compatible when switching from terrestrial powders to ISRU-derived inputs based on mare-type basalt (Fe–Ti–rich) or highlands-type anorthosite (Al–Ca–rich). Concepts for blending ISRU metals (Al, Ti, Fe, Mg, Ca from regolith and debris) with refractory bases to produce graded components that combine high-temperature strength, oxidation resistance and, where needed, radiation shielding or KREEP-derived functional layers. A digital manufacturing toolkit (DoE + surrogate models) to predict density, defects and microstructure from process parameters, reducing empirical trials when adapting to different terranes (mare, highlands, PKT / South Pole–Aitken rim). We present current SPARK data (TRL 3–4) together with a TRL roadmap toward a modular metallurgical station for the Moon, aligned with ESA PhiLab and E3P priorities and designed to close material loops rather than ship every kilogram from Earth.

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1 Introduction

The return to the Moon under NASA's Artemis programme and ESA's European Exploration Envelope Programme (E3P) envisions sustained human presence beyond short-duration sortie missions. A permanent lunar outpost will require structural metals, thermal-management components, and radiation shielding in quantities that make full Earth-supply economically prohibitive. In-Situ Resource Utilization (ISRU) for oxygen production from regolith has reached laboratory maturity,carbothermal reactors and molten regolith electrolysis (MRE) cells have been demonstrated at the kilogram scale [1, 2],yet the downstream question of how to turn extracted metals into qualified engineering alloys and components remains largely unaddressed. Closing this gap requires coupling real geochemical resource maps to metallurgical process design from the outset.

The lunar crust is not a single homogeneous feedstock. Jolliff et al. [3] established a three-terrane geochemical framework based on Lunar Prospector Gamma Ray Spectrometer (LP-GRS) data [4]: (i) feldspathic highlands, dominated by plagioclase and rich in Al_2O_3 and CaO; (ii) mare basaltic regions, enriched in FeO and TiO_2 with abundant ilmenite ($FeTiO_3$); and (iii) the Procellarum KREEP Terrane (PKT), enriched in incompatible elements (K, rare-earth elements, Th, U). Each terrane presents a distinct extraction chemistry and therefore a different starting point for alloy design [5, 6]. Treating all regolith as equivalent,the implicit assumption in most ISRU architecture studies,introduces systematic errors in feedstock quality estimates and downstream process sizing.

Refractory high-entropy alloys (RHEAs) have emerged as a frontier material class for extreme-temperature structural applications. Body-centred cubic RHEAs based on refractory elements (Mo, Nb, Ta, W, V, Hf, Ti, Zr, Cr) retain useful strength well above the service ceiling of conventional nickel superalloys, which suffer pronounced property degradation above $\sim$700°C [7–9]. Within ESA's FIRST! initiative, part of the Future Launchers Preparatory Programme (FLPP), the SPARK project (Strong Performance Alloys for Rocket Kinetics) has screened eleven candidate compositions across two alloy families –five-element refractory HEAs (SPARK-R series) and eight-element high-entropy superalloys (SPARK-H series), together with a proprietary Ni-based superalloy (SPARK-S1) –for oxygen-rich preburner environments in staged-combustion rocket engines, where operating temperatures exceed 1 000 K and pressures reach up to 380 bar [10].[1] These alloys are synthesized via mechanical alloying (MA, 40 h in heptane process-control agent) followed by vacuum hot pressing

[1]Exact alloy compositions are withheld under the terms of the ESA FIRST! consortium agreement. Compositions are identified by programme designators (SPARK-R1 through SPARK-R6 for five-element refractory HEAs, SPARK-H1 through SPARK-H3 for eight-element high-entropy superalloys, and SPARK-S1 for a proprietary Ni-based superalloy). Processing parameters, phase identifications, and mechanical data are reported in full to enable process-window assessment without disclosing proprietary formulations.

(HP, 1 500°C / 2 h / 30 MPa) for rapid composition screening and densification, with transfer to laser powder-bed fusion (LPBF) and directed energy deposition (DED) planned for near-net-shape manufacture of down-selected candidates. Mechanical characterisation (mini-tensile tests at room temperature and 1 000 K, Brinell hardness at RT and 1 000 K, Vickers micro-hardness HV2) and phase analysis (XRD at each processing stage; EDS elemental mapping of powders, compacts, and remelted ingots) provide the screening data. Oxidising gas-flow testing under ERBURIK conditions (∼1 000 K) at ArianeGroup is planned for down-selected candidates. The Earth-side process chain established in SPARK,from MA/HP screening through planned LPBF/DED manufacture to hot-section testing,provides a calibrated baseline from which lunar feedstock adaptation can proceed.

Current ISRU metal-extraction research concentrates on three routes: hydrogen reduction of ilmenite for Fe and O_2 [11]; molten regolith electrolysis yielding Fe–Si alloys and oxygen [2]; and molten salt electrolysis (FFC-Cambridge process) producing Al–Si alloys. Additive manufacturing of lunar regolith has been explored primarily for civil-engineering applications (sintered habitats, landing pads), but metal additive manufacturing from ISRU-derived feedstock remains largely unexplored. No published framework connects terrane-resolved geochemistry to extracted-metal quality, alloy-family selection, process-window definition (HP, LPBF, DED), and qualification logic as a single integrated chain. This is the gap that the present work addresses.

We present a terrane-aware metallurgical framework that maps LP-GRS 2° geochemical data through compositional data analysis (CLR transform and PCA) into three metallurgical potential indices, and links these to candidate alloy families (RHEA, HEA, Ni-superalloy) and manufacturing processes via a compatibility matrix informed by SPARK Earth-side trials. The key novelty is the integration of:

- terrane chemistry stratification from orbital spectroscopy,
- extraction-to-alloy feasibility mapping across three resource classes,
- process-window transfer logic anchored in SPARK experimental baselines (MA/HP screening → LPBF/DED manufacture), and
- a qualification roadmap that identifies the experimental campaigns needed when adapting to different lunar feedstocks.

Key Highlight

This manuscript presents a **transfer framework anchored in SPARK experimental data**, not a completed lunar industrial implementation. SPARK screening (MA + HP, 9 compositions, XRD/EDS/mechanical testing) is complete; down-selection to two candidates is reported. Exact compositions are withheld under the consortium IP agreement; processing parameters and properties are reported in full. Microstructural characterisation of down-selected alloys is in progress.

2 Scope, Claim Boundary, and Evidence Classes

This paper presents a *framework* connecting orbital geochemistry to alloy process-window selection, not a demonstrated lunar manufacturing capability. To maintain technical credibility, every substantive claim is classified into one of four evidence tiers: **Measured** (Earth-side data from published sources or SPARK trials), **Model-derived** (computed from measured data via standard methods), **Assumed** (expert judgement requiring future validation), and **Roadmap hypothesis** (no experimental demonstration yet). Table 8 maps the paper's key assertions to these tiers.

3 Data Sources and Analytical Workflow

3.1 Terrane-oriented geochemical dataset

Our primary data source is the Lunar Prospector Gamma Ray Spectrometer (LP-GRS) global elemental abundance dataset at $2^\circ \times 2^\circ$ equal-area spatial resolution, compiled by Prettyman et al. [4]. This product provides pixel-level concentrations for six major oxides and three trace elements derived from orbital gamma-ray spectroscopy.

Table 1: LP-GRS 2-degree dataset schema. Major oxides are reported in weight percent (wt%); trace elements in parts per million (ppm). The U abundance is tied to Th via the empirical relation $\mathrm{U} = 0.27\,\mathrm{Th}$ used in the LP-GRS unmixing model [4] and is therefore excluded from PCA to avoid introducing a linear dependency.

Column	Unit	Role in pipeline	Transform
lat, lon	deg	spatial bin identifier	—
FeO	wt%	PCA feature + I_{FeTi}	CLR
TiO_2	wt%	PCA feature + I_{FeTi}	CLR
Al_2O_3	wt%	PCA feature + I_{AlCa}	CLR
CaO	wt%	PCA feature + I_{AlCa}	CLR
MgO	wt%	PCA feature	CLR
SiO_2	wt%	PCA feature + penalty	CLR
K	ppm	PCA feature + I_{KREEP}	z-score
Th	ppm	PCA feature + I_{KREEP} + classifier	z-score
U	ppm	I_{KREEP} only	z-score (not in PCA)
terrane	–	Jolliff classification	—

The global dataset comprises **11 306 spatial pixels** covering the near-complete lunar surface. For metallurgical planning we define three operational resource classes broadly aligned with the geochemical framework of Jolliff et al. [3], who identified the Feldspathic Highlands Terrane (FHT), the Procellarum KREEP Terrane (PKT), and the South Pole–Aitken Terrane (SPAT). Our simplified scheme merges SPAT into the highlands class and separates mare basaltic units from KREEP-bearing regions using the following thresholds:

- **Highlands** ($n = 8{,}472$; 74.9%): Th $\leq$ 3.5 ppm and (FeO $\leq$ 8 wt% or $TiO_2 \leq$ 0.5 wt%).
- **Mare** ($n = 1{,}222$; 10.8%): FeO $>$ 8 wt% and $TiO_2 >$ 0.5 wt%.
- **KREEP-bearing** ($n = 1{,}612$; 14.3%): Th $>$ 3.5 ppm.

The dominance of highlands pixels reflects the lunar far side, which is almost entirely feldspathic crust [5].

3.2 Compositional preprocessing and PCA

The six major oxides obey a constant-sum constraint (∼97 wt%), so we apply the Centered Log-Ratio (CLR) transform [12] to address the closure problem [13]:

$$\mathrm{clr}(c_i) = \ln\left(\frac{c_i}{g(\mathbf{c})}\right), \quad g(\mathbf{c}) = \left(\prod_{j=1}^{D} c_j\right)^{1/D}, \tag{1}$$

where $D = 6$. The trace elements K and Th are z-scored independently and appended to the CLR features, creating a hybrid metric space (log-ratio + Euclidean); CLR is retained over ILR [14] for interpretability. U is excluded from PCA because the LP-GRS unmixing model ties it to Th (U $= 0.27\,$Th) [4].

Data cleaning. Unmixing artifacts are clipped to physically motivated bounds (FeO 0–28, TiO_2 0–15, Al_2O_3 0–38, CaO 0–22, MgO 0–25, SiO_2 35–55 wt%) and replaced via terrane-wise median imputation (1 414 values clipped). Below-detection zeros (2 859 total, predominantly highlands TiO_2) are replaced with terrane-wise medians rather than epsilon-clipping, which would create $\sim\!-21\sigma$ outliers in CLR space. Approximately 38% of pixels have at least one modified value.

PCA. PCA [15] is applied to the $11{,}306 \times 8$ feature matrix. Loadings are stabilised via 200 sign-aligned bootstrap resamples [16] ($\sigma \leq 0.034$, median $\sigma = 0.014$). Terrane separability is quantified by the silhouette coefficient [17] (0.300, "weak but detectable" per Kaufman–Rousseeuw 1990) and the Davies–Bouldin Index [18] (1.247). These metrics confirm that the three operational resource classes are statistically distinguishable in PCA space, though they overlap along continuous geochemical gradients as expected at 2° resolution.

3.3 Terrane Metallurgical Potential Indices

To translate geochemical composition into metallurgical relevance, we define three terrane-specific indices. All oxide and trace-element concentrations are first min-max normalized to $[0, 1]$ over the full dataset before weighting.

Fe–Ti index (mare basalt feedstock potential):

$$I_{\text{FeTi}} = 0.50\,\hat{c}_{\text{FeO}} + 0.40\,\hat{c}_{\text{TiO}_2} - 0.10\,P_{\text{imp}}, \tag{2}$$

where $P_{\text{imp}} = 0.6\,\hat{c}_K + 0.4\,\hat{c}_{\text{Th}}$ penalizes KREEP contamination in otherwise Fe–Ti-dominated pixels. High I_{FeTi} identifies regions suitable for ilmenite-based iron/titanium extraction.

Al–Ca index (highlands structural metal potential):

$$I_{\text{AlCa}} = 0.55\,\hat{c}_{\text{Al}_2\text{O}_3} + 0.35\,\hat{c}_{\text{CaO}} - 0.10\,P_{\text{het}}, \tag{3}$$

where P_{het} is the min-max-normalized absolute deviation of SiO_2 from its terrane mean, serving as a proxy for compositional mixing. High I_{AlCa} identifies feldspathic regions suitable for aluminum and calcium extraction.

KREEP index (incompatible-element functional potential):

$$I_{\text{KREEP}} = 0.40\,\hat{c}_K + 0.40\,\hat{c}_{\text{Th}} + 0.10\,\hat{c}_U - 0.10\,P_{\text{unc}}, \tag{4}$$

where $P_{\text{unc}} = |\text{lat}|$ (min-max normalized) is a proxy for LP-GRS counting-statistics quality (polar orbits have fewer accumulated counts). Note that U is included in I_{KREEP} (where its physical role as a heat-producing element is relevant) even though it is excluded from PCA (where its linear dependence on Th would be a statistical artifact). High I_{KREEP} flags regions with enriched incompatible elements relevant for functional layers and radiation shielding.

In the above, $\hat{c}_x$ denotes the min-max-normalized concentration of species x, scaled to $[0, 1]$ over the full dataset. All indices are clipped to $[0, 1]$ after computation. Note that the positive weights in each index sum to 0.90 while the penalty weight is 0.10, so the theoretical maximum (all positive-weight species at maximum, zero penalty) is 0.90; the upper clip at 1.0 is never binding.

3.4 Alloy-family portfolio mapping

The alloy-family portfolio mapping takes terrane chemistry (x_t), feedstock quality (q_f), and component duty (d_c) as inputs, and returns a candidate alloy family (a_f), blend window (b_w), process route (p_r), and confidence level (c_ℓ):

$$\mathcal{F}(x_t,\, q_f,\, d_c) \;\rightarrow\; (a_f,\, b_w,\, p_r,\, c_\ell).$$

This is a schematic mapping, not a closed-form equation; the decision logic is implemented as the compatibility matrix in Table 10.

4 Results

4.1 PCA terrane separation

The PC1–PC2 projection (Figure 1) shows the highlands cluster as compact and well-separated, with mare and KREEP overlapping along PC1. The first three principal components capture **83.6%** of total variance (Table 2). Loadings and their bootstrap standard deviations are listed in Table 9.

Table 2: Explained variance for the first three principal components.

	Explained variance (%)	Cumulative (%)
PC1	53.1	53.1
PC2	17.4	70.4
PC3	13.2	83.6

Geochemical interpretation of principal components. **PC1** (53.1% variance) captures the fundamental **highlands vs. mare/KREEP dichotomy**,the dominant gradient in lunar crustal geochemistry. The negative loadings on Al_2O_3 (-0.39), SiO_2 (-0.40), and CaO (-0.35) place the feldspathic highlands at the negative end of this axis, reflecting the anorthositic (plagioclase-rich) composition of the ancient lunar crust. The positive loadings on FeO ($+0.31$), TiO_2 ($+0.37$), K ($+0.40$), and Th ($+0.42$) place mare basalts and KREEP-bearing units at the positive end, reflecting their enrichment in iron-group and incompatible elements relative to the highlands. This component rediscovers the well-established geochemical bimodality of the lunar crust [3, 5].

PC2 (17.4% variance) is dominated by **MgO** ($+0.74$), with secondary negative loadings on CaO (-0.39), FeO (-0.35), and SiO_2 (-0.31). This axis separates mafic (Mg-rich pyroxene/olivine-bearing) lithologies from more calcic or ferroan compositions. Geologically, this distinguishes Mg-suite intrusive rocks and magnesian mare basalts from ferroan anorthosites and high-Ti mare basalts.

PC3 (13.2% variance) is dominated by **TiO_2** ($+0.61$) with opposing MgO (-0.44) and FeO (-0.39). This axis captures the well-known **high-Ti vs. low-Ti mare basalt** distinction [6]: high-Ti basalts (e.g. Apollo 11 and 17 sites) are enriched in ilmenite, whereas low-Ti basalts (e.g. Apollo 12 and 15 sites) have higher MgO. This distinction is directly relevant to metallurgy because high-Ti mare pixels are the primary targets for ilmenite-based oxygen and iron extraction.

The loading vectors are shown in Figure 2.

Nonlinear validation via UMAP. To confirm that the terrane separation observed in PCA is not an artifact of the linear projection, we computed a UMAP embedding (Uniform Manifold Approximation and Projection; $n_{\text{neighbors}} = 30$, $d_{\text{min}} = 0.3$) of the same 8-dimensional standardised feature space. Figure 3 shows the resulting two-dimensional manifold coloured by terrane class (left) and by I_{FeTi} (right). The terranes form a continuous manifold rather than discrete clusters, with highlands occupying the dominant mass, mare as a distinct arm, and KREEP bridging the two. The I_{FeTi} gradient maps smoothly onto this structure, confirming that the index captures genuine geochemical variation rather than threshold artifacts. Separability metrics in UMAP space (silhouette = 0.291, DBI = 1.089) are consistent with those from PCA (0.300, 1.247), confirming that the moderate separation reflects real geochemical gradients rather than a projection artifact of either method.

4.2 Manufacturability-access mapping

The three metallurgical potential indices are projected back onto selenographic coordinates to produce global resource maps (Figures 4–5). Terrane-level summary statistics are given in Table 3.

Table 3: Metallurgical potential indices by terrane (mean $\pm$ 1 s.d.). The indices are dimensionless, clipped to $[0, 1]$ with a theoretical maximum of 0.90 (positive weights sum to 0.90 in all three indices).

	I_{FeTi}	I_{AlCa}	I_{KREEP}
Highlands	0.104 ± 0.034	0.626 ± 0.076	0.014 ± 0.026
Mare	0.490 ± 0.109	0.186 ± 0.078	0.096 ± 0.044
KREEP-bearing	0.307 ± 0.104	0.225 ± 0.088	0.385 ± 0.106

The indices confirm the expected terrane–resource associations:

- **Highlands** pixels score highest on I_{AlCa} (0.626 ± 0.076) and lowest on both I_{FeTi} and I_{KREEP}, consistent with their anorthositic composition.
- **Mare** pixels lead on I_{FeTi} (0.490 ± 0.109), reflecting their iron- and titanium-rich basaltic fill.
- **KREEP-bearing** pixels have the highest I_{KREEP} (0.385 ± 0.106) and moderate I_{FeTi} (0.307 ± 0.104), because the Procellarum KREEP Terrane is geographically co-located with several major

mare basins.

4.3 Sensitivity of metallurgical index weights

The metallurgical potential indices (Eqs. 2–4) incorporate expert-assigned weights that have not been formally optimised. To assess the robustness of terrane rankings to weight uncertainty, we performed a Monte Carlo sensitivity analysis with $n = 10{,}000$ parameter perturbations.

Perturbation protocol. For each iteration, every nominal weight w_i was perturbed by drawing from a truncated normal distribution:

$$w_i' = w_i \cdot (1 + \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad |\epsilon_i| \leq 0.5, \tag{5}$$

where $\sigma = 0.2$ (i.e., $\pm 20\%$ standard deviation, truncated at $\pm 50\%$). After perturbation, the positive weights within each index were renormalised to preserve their nominal sum while the penalty-term magnitudes were allowed to float. For each realisation, the three indices were recomputed for all 11 306 pixels and terrane-mean values recorded.

Index distributions. Figure 9 shows the distribution of terrane-mean index values across the 10 000 Monte Carlo realisations. Table 4 reports the 5th, 50th, and 95th percentiles.

Table 4: Monte Carlo sensitivity results: terrane-mean index values (5th / 50th / 95th percentiles) across 10 000 weight perturbations ($\sigma = 0.20$, truncated at $\pm 50\%$).

Index	Highlands	Mare	KREEP-bearing
I_{FeTi}	0.084 / 0.104 / 0.124	0.448 / 0.490 / 0.531	0.273 / 0.307 / 0.341
I_{AlCa}	0.618 / 0.626 / 0.634	0.172 / 0.186 / 0.201	0.214 / 0.225 / 0.236
I_{KREEP}	0.011 / 0.014 / 0.020	0.087 / 0.096 / 0.105	0.370 / 0.385 / 0.400

Ranking stability. Across all 10 000 realisations, the terrane rankings are perfectly stable: mare pixels maintain the highest mean I_{FeTi} in 100% of cases, highlands pixels lead on I_{AlCa} in 100% of cases, and KREEP-bearing pixels dominate I_{KREEP} in 100% of cases. No ranking flip occurred under any sampled weight perturbation.

The 90% confidence intervals (5th–95th percentile range) are narrow relative to the inter-terrane gaps. For example, the I_{FeTi} 95th percentile for KREEP (0.341) falls well below the 5th percentile for Mare (0.448), confirming that the separation is robust even under worst-case weight perturbation.

Among the three indices, I_{AlCa} shows the tightest distributions ($\pm 1\%$ of the median), while I_{FeTi} and I_{KREEP} show moderate spread (± 8–15% of the median).

Per-weight sensitivity. Figure 10 maps the effect of individual weight perturbations on inter-terrane separation for each index. The dominant weights (w_{FeO}, $w_{\text{Al}_2\text{O}_3}$, w_{K}) drive up to $\pm 25\%$ change in terrane separation at $1.5\times$ perturbation, while penalty terms (P_{imp}, P_{het}, P_{unc}) have near-zero effect. This confirms that the rankings are driven by genuine geochemical contrasts rather than by the penalty calibration.

4.4 Terrane-to-alloy/process compatibility

Table 10 maps each terrane chemistry to candidate alloy families and manufacturing processes, with associated risks and evidence classes.

4.5 Published process baselines for candidate alloy families

The published literature provides calibration baselines for the processing conditions and elevated-temperature performance of RHEA systems produced by SPS. Table 11 compiles SPS and HP process windows from the literature and from the SPARK campaign for representative RHEA compositions, and Table 12 compares their tensile strength at elevated temperatures against the Ni-superalloy benchmarks.

Key observations: SPS temperatures for refractory-heavy compositions (Mo, Nb, Ta, W dominant) cluster in the 1 500–1 800°C range at 50–80 MPa, with hold times of 2–20 min. Alloys containing lower-melting constituents (Hf, Ti, Zr) can be sintered at lower temperatures ($\sim$1 300°C). MA + SPS consistently produces finer grain sizes and higher yield strength than arc-melting for the same composition [19]. Illustrative phase-fraction calculations for two reference compositions (equimolar HfNbTaTiZr and MoNbTaTiV), computed via COST507 extrapolation, suggest single-phase BCC stability across the solid-state range (Figure 7a,b); however, COST507 lacks binary interaction parameters for most refractory pairs, so these predictions should be treated as indicative rather than confirmatory; experimental XRD from SPARK trials and published literature [19, 20] provides the primary evidence for BCC phase stability.

The ISRU column quantifies the blending opportunity using a conservative element set: only Fe, Ti, Al, Si, and Mg are counted as lunar-extractable, since these have demonstrated ISRU extraction routes (H_2 reduction, MRE, FFC-Cambridge). Elements present in trace lunar minerals but lacking demonstrated extraction processes (e.g. Zr, Cr) are counted as Earth-supplied.

For equimolar HfNbTaTiZr, only Ti is lunar-sourced ($\sim$8 wt%). For AlMo$_{0.5}$NbTa$_{0.5}$TiZr,the most promising published RHEA for the 800–1 000°C regime (2 000 MPa at RT, 745 MPa at 1 000°C, density 7.4 g/cm^3 [7]),Al and Ti together provide ∼19 wt% from ISRU. Alloys composed entirely of heavy refractories (MoNbTaW, MoNbTaVW) have 0% lunar extractability and must be shipped whole. Notably, Inconel 718 has ∼20% ISRU potential (its Fe and minor Ti/Al content), but the dominant Ni and Cr fractions remain Earth-dependent.

These modest single-digit to ∼20% ISRU fractions for *existing* alloy compositions underscore the paper's central argument: the primary value of process-window calibration (SPARK) is not to reproduce these exact compositions on the Moon, but to establish the HP/SPS parameter space within which **new blends**,dominated by ISRU-derived Fe–Ti or Al–Si, supplemented with smaller Earth-shipped refractory additions,can be explored with predictive confidence rather than trial-and-error.

Published data compiled from the Senkov–Miracle mechanical property database [7, 20, 21]. FFC-Cambridge outputs characterised by Lomax et al. [22].

4.6 SPARK experimental characterisation

Eleven compositions were modelled using Thermo-Calc CALPHAD: six five-element refractory HEA variants (SPARK-R1 through SPARK-R6, differentiated by a single additive element to a common refractory base) and three eight-element high-entropy superalloys (SPARK-H1 through SPARK-H3). A proprietary Ni-based superalloy (SPARK-S1), identified in a partner project, was included as a comparison. Nine compositions were experimentally produced via mechanical alloying (MA, 40 h, heptane PCA) followed by vacuum hot pressing (HP, 1 500°C / 2 h / 30 MPa, except SPARK-S1 at 1 260°C). Arc remelting was performed on selected compositions to assess melt behaviour and segregation.

Phase analysis. XRD confirmed the computationally predicted phases at each processing stage. SPARK-R1 exhibited a dominant single-phase BCC structure from MA through HP and arc remelting. The eight-element HESA compositions showed more complex phase assemblages: SPARK-H1 and SPARK-H3 exhibited BCC + MC carbide phases, while SPARK-S1 showed FCC + MC phases after HP. XRD of SPARK-H2 after heating to 1 000°C showed significant peak multiplication relative to the as-HP pattern, indicating phase decomposition that may contribute to embrittlement.

EDS elemental mapping (Figure 8) confirmed homogeneous distribution of all five constituents in SPARK-R1 across both HP and remelted conditions, consistent with the single-phase BCC structure. In contrast, SPARK-R3 exhibited clear segregation of one constituent to grain boundaries (Figure 8b), producing a two-phase microstructure visible in backscatter SEM. The remelted HESA ingots showed dendritic segregation with refractory-element-rich cores and lighter-element-rich interdendritic regions.

Mechanical properties. Tables 11 and 12 include the SPARK results alongside published baselines. Two results are noteworthy:

(i) SPARK-R1, a five-element BCC RHEA, exhibits an anomalous strength increase at 1 000 K: UTS rises from ∼504 MPa at room temperature to ∼532 MPa at 1 000 K ($+5.6$%), accompanied by Brinell hardness of 1 078 HB at RT. This intermediate-temperature strengthening is characteristic of thermally activated dislocation pinning in BCC refractory HEAs [7] and is directly relevant to the preburner operating regime (1 000–1 200 K).

(ii) All three eight-element HESA compositions (SPARK-H1, H2, H3), despite excellent hardness (500–840 HV2) and predicted thermal stability, exhibited extreme brittleness during all processing stages. Samples fractured during demoulding; mini-tensile specimens either could not be tested or returned UTS values below 170 MPa with negligible plastic deformation. The five-element RHEA variants SPARK-R5 and SPARK-R6 also proved brittle, with UTS of 171 and 191 MPa respectively.

Down-selection. Two candidates were selected for further development toward combustion-chamber applications: SPARK-R1, for its single-phase BCC stability, oxidation-resistant additive, and anomalous high-temperature strengthening; and SPARK-S1, a proprietary Ni-based superalloy with 1 250 MPa RT UTS and 510 MPa at 1 000 K, combining high strength with improved ductility over the refractory HESAs.

Comparison with Inconel benchmark. An Inconel specimen produced via the same HP route returned ∼738 MPa (RT) and ∼974 MPa (1 000 K) under identical mini-tensile geometry. The apparent strength increase at temperature requires cautious interpretation: it may reflect precipitation hardening (γ'' in the 600–800°C window), specimen geometry effects, or both. At 1 000 K, Inconel (∼974 MPa) outperforms SPARK-R1 (∼532 MPa); however, the SPARK-R1 trend –maintaining or gaining strength at temperature –suggests a performance crossover at higher temperatures where Ni-superalloy degradation accelerates. Confirming this crossover requires tensile data at 1 200–1 500 K, planned for the next campaign.

5 Discussion

The results support terrane-specific manufacturing logic rather than a one-size-fits-all approach. This section traces the logic chain from geochemical prediction through process calibration to the blending concept that connects ISRU bulk metals with SPARK-qualified refractory alloy routes.

The silhouette coefficient of 0.300 should be understood in context: the terrane boundaries are threshold-based cuts on a continuous geochemical gradient, not natural clusters. Perfect separation would be suspicious given the spatial resolution (2° pixels blend multiple lithologies) and the gradational nature of terrane transitions. The value falls in the 0.25–0.50 range conventionally described as "weak structure," yet the separability is sufficient to assign distinct metallurgical strategies with quantified confidence.

5.1 What the PCA tells us: feedstock prediction from orbit

The three principal components recovered from LP-GRS data rediscover the established geochemical gradients of the lunar crust,highlands vs. mare enrichment, mafic differentiation, and Ti variability,without prior knowledge of sample-return petrology, demonstrating that orbital spectroscopy alone provides sufficient chemical resolution for first-order metallurgical planning. The metallurgical potential indices translate these gradients into actionable resource predictions:

- **Mare pixels** (I_{FeTi} dominant) identify regions where hydrogen reduction of ilmenite or carbothermal processing can yield Fe–Ti feedstock and process oxygen [1, 11].
- **Highlands pixels** (I_{AlCa} dominant) flag anorthositic crust suited to molten regolith electrolysis or the FFC-Cambridge process, producing Al–Si structural alloys.
- **KREEP-bearing pixels** present a dual opportunity: co-located I_{FeTi} and I_{KREEP} in the Procellarum region (Table 3) enable combined Fe–Ti extraction with recovery of high-Z elements (Th, U) for radiation-shielding applications.

Crucially, these ISRU routes produce *bulk metals* (Fe, Ti, Al, Si, Mg, and oxygen), not the refractory elements (Mo, Nb, Ta, W, Hf, V, Cr) that constitute RHEA compositions. The Moon's crust is depleted in siderophile and highly refractory elements relative to Earth; no known lunar extraction process yields Mo, Nb, or Ta at useful purity. This asymmetry is central to the framework's logic.

5.2 What SPARK provides: calibrated process knowledge

The SPARK project screened eleven compositions across two alloy families for oxygen-rich preburner components in staged-combustion rocket engines, where the benchmark material (Inconel / Monel K500) suffers property degradation above ∼700°C. The experimental campaign comprised mechanical alloying (MA, 40 h, heptane PCA), vacuum hot pressing (HP, 1 500°C / 2 h / 30 MPa), and arc remelting, with phase analysis (XRD, EDS) at each stage and mechanical testing (mini-tensile at RT and 1 000 K, Vickers and Brinell hardness) on the densified compacts.

The campaign produced two principal findings. First, SPARK-R1, a five-element BCC RHEA, demonstrates single-phase BCC stability across all processing stages, homogeneous microstructure after hot pressing, and anomalous strength retention at 1 000 K (532 MPa, a 5.6% increase over the RT value of 504 MPa). Second, the eight-element HESA compositions –despite excellent hardness (500–840 HV2) and computational predictions of thermal stability –are too brittle for structural service, fracturing during demoulding and returning negligible ductility under tension.

For the lunar transfer framework, SPARK's deliverable is a **calibrated process chain**: powder morphology after 40 h MA (sub-micron, confirmed by SEM), HP densification profiles (1 500°C / 30 MPa / 2 h achieving 9.4–12.9 g/cm^3 across compositions), phase stability maps validated against XRD, and process–property relationships linking input chemistry to density, phase assemblage, hardness, and tensile response. Transfer to LPBF and DED for near-net-shape manufacture of SPARK-R1 and SPARK-S1 is the next programme milestone. The HP windows established here define the sintering parameter space within which input powder chemistry can be varied –including substitution of ISRU-derived Fe–Ti or Al–Si for a fraction of the refractory elements –while maintaining predictable output quality.

5.3 The bridge: blending ISRU bulk metals with Earth-shipped refractories

The connection between ISRU feedstock prediction (Section 5.1) and SPARK process knowledge (Section 5.2) operates through a **blending concept**.

The SPARK experimental results provide a materials-science motivation for the blending concept beyond mass savings. The eight-element refractory HESAs (SPARK-H1 through H3) exhibit hardness values of 500–840 HV2 –among the highest reported for bulk HEA compacts –but are catastrophically brittle, rendering them unsuitable for structural applications as monolithic components. Alloying with ISRU-derived ductile-phase metals (Fe, Ti, Al) offers a potential route to improve toughness: Fe–Ti from mare ilmenite reduction could serve as a ductile BCC matrix into which small fractions of refractory elements are dissolved or dispersed as strengthening phases, analogous to the oxide-dispersion-strengthened (ODS) steel concept but using HEA strengtheners rather than oxide particles. The SPARK brittleness data thus reframes the blending question from "how much Earth mass can we save?" to "can ISRU-derived ductile phases solve the toughness problem that limits monolithic refractory HESAs?"

The blending logic proceeds in three steps:

1. ISRU extraction at a given terrane produces bulk metal powder (e.g. Fe–Ti from mare ilmenite, Al–Si from highlands anorthosite).
2. This ISRU-derived powder is blended with small quantities of Earth-shipped refractory elements (Mo, Nb, Ta, W) to form a feedstock compatible with the SPARK-calibrated process chain.
3. The blended feedstock undergoes HP screening to verify sinterability and phase stability, followed

by LPBF or DED manufacture of near-net-shape components using SPARK-calibrated build parameters. The process–property relationships from SPARK predict whether acceptable density and microstructure will be achieved, reducing the experimental campaign needed for each new composition.

In early lunar operations, the refractory fraction is entirely Earth-supplied,but because refractories constitute a minor mass fraction of the final component (typically 5–30 wt% depending on the alloy system and duty), the total Earth-import burden is sharply reduced compared to shipping finished parts. A mare-derived Fe–Ti powder blended with ∼10 wt% Earth-supplied Mo and Nb, screened via HP and then manufactured by LPBF or DED through SPARK-calibrated process windows, could yield a structural alloy with high-temperature capability exceeding that of pure ISRU iron,while importing only a tenth of the component mass from Earth.

An important caveat applies to the blending concept: adding lower-melting ISRU-derived elements (Al, Si, and to a lesser extent Ti) to refractory bases (Mo, Nb, Ta, W) depresses the solidus temperature, potentially compromising the high-temperature strength that motivates RHEA selection. This trade-off suggests a **graded architecture** for lunar components: ISRU-dominated alloys (Fe–Ti, Al–Si) for structural applications below ∼800°C, blended compositions (ISRU base + 10–30 wt% Earth-shipped refractories) for intermediate-temperature service, and Earth-pure RHEAs reserved for the most demanding hot-section components (combustion chambers, nozzle throats) where temperatures exceed 1 200°C. The binary phase diagrams for the Fe–Ti, Al–Ti, Al–Fe and Al–Ni systems (Figure 6) illustrate the phase fields accessible from ISRU feedstocks and help bound the compositional space for each service tier.

As ISRU extraction technology matures and diversifies beyond the first-generation Fe/Ti/Al/Si routes, the Earth-shipped fraction can progressively shrink. The metallurgical potential indices provide the planning tool: they identify which terranes offer which bulk metals, enabling mission architects to size the refractory supplement inventory against the local resource base.

5.4 Broader implications

The framework presented here does not claim to solve lunar manufacturing; rather, it provides a structured methodology for deciding *where on the Moon to extract what*, *which bulk metals become available*, and *how to combine them with Earth-supplied elements* using process windows calibrated in SPARK. This terrane-aware approach reduces the combinatorial search space for future ISRU demonstration missions and provides quantitative site-selection metrics that complement terrain trafficability and volatile-access considerations in landing-site trade studies. Once SPARK trials yield validated process–property relationships across the HP→LPBF/DED chain, these relationships can inform how shifting from terrestrial to blended ISRU/refractory powder chemistry will affect density, defect populations, and microstructure,reducing the number of physical experiments needed during proxy-feedstock demonstration campaigns.

6 Economic Considerations

The terrane-aware metallurgy framework has direct implications for lunar infrastructure economics. This section develops a parametric cost model that maps the decision space for ISRU-integrated manufacturing without claiming precision on inherently uncertain future costs. All economic projections are classified as **Roadmap hypothesis** (Tier R) per Table 8.

6.1 ISRU Mass Leverage Ratio

We define the *ISRU Mass Leverage Ratio* (IMLR) as the ratio of final component mass to Earth-shipped feedstock mass:

$$\text{IMLR} = \frac{M_{\text{component}}}{M_{\text{Earth}}} = \frac{M_{\text{ISRU}} + M_{\text{Earth}}}{M_{\text{Earth}}}, \tag{6}$$

where M_{ISRU} is the mass contribution from lunar-extracted metals (Fe, Ti, Al, Si, Mg) and M_{Earth} is the mass of Earth-shipped feedstock (refractory elements Mo, Nb, Ta, W, Hf, V, Cr plus any required alloying additions not extractable on the Moon).

For the candidate alloy systems in Table 12, IMLR ranges from 1.0 (pure Earth-shipped RHEAs like MoNbTaW) to approximately 1.25 ($AlMo_{0.5}NbTa_{0.5}TiZr$ with 19% ISRU content). The blending concept proposed in Section 5.3 targets IMLR values of 5–10 for structural applications, achieved by using ISRU-derived Fe–Ti or Al–Si as the matrix phase with 10–20 wt% Earth-shipped refractory additions.

6.2 Parametric Cost Model

The total cost C_{total} to produce N components of mass $M_{\text{component}}$ via ISRU-integrated manufacturing comprises launch costs, capital expenditure, and operating costs:

$$C_{\text{total}} = \underbrace{C_L \cdot M_{\text{Earth}} \cdot N}_{\text{Launch}} + \underbrace{C_{\text{cap}}}_{\text{Capital}} + \underbrace{C_{\text{op}} \cdot M_{\text{ISRU}} \cdot N}_{\text{Operations}}, \tag{7}$$

where C_L is the cost to deliver 1 kg to the lunar surface (\$/kg), C_{cap} is the capital cost of the ISRU metallurgical station including launch mass (\$), C_{op} is the operating cost to extract and process 1 kg of ISRU metal (\$/kg), and N is the number of components produced over the station lifetime.

The break-even condition against pure Earth-supply ($C^{\text{ISRU}}_{\text{total}} = C^{\text{Earth}}_{\text{total}}$, where $C^{\text{Earth}}_{\text{total}} = C_L \cdot M_{\text{comp}} \cdot N$)

yields, after substituting $M_{\text{Earth}} = M_{\text{comp}}/\text{IMLR}$:

$$C_{\text{op}} < C_L - \frac{C_{\text{cap}}}{N \cdot M_{\text{ISRU}}}. \tag{8}$$

This inequality defines the *viable parameter space* for ISRU-integrated manufacturing: the per-kilogram operating cost of ISRU extraction must be less than the launch cost minus the amortised capital burden. Notably, break-even depends on production volume N: higher throughput amortises capital costs and expands the viable operating cost range.

6.3 Launch Cost Scenarios

Table 5 summarises lunar surface delivery costs across launch architectures, grounded in publicly available contract and pricing data. The wide range,spanning two orders of magnitude,reflects the rapid evolution of the launch market.

Table 5: Lunar surface delivery cost scenarios (2026 baseline). CLPS costs are derived from NASA contract values divided by payload capacity [23]. The Falcon Heavy LEO cost of \$1 400/kg [24] is scaled by a $\sim$7× mass multiplier for trans-lunar injection, orbit insertion, and soft landing. Starship projections assume 100 t cargo to lunar surface; fully reusable targets per Jones [24].

Scenario	**Architecture**	C_L **(\$/kg)**	**Status**
CLPS (small lander)	IM-1 Nova-C (\$118M / $\sim$100 kg)	$\sim$1 200 000	Demonstrated (2024)
Heavy lander	FH + Griffin-class (>500 kg)	$\sim$200 000	Near-term
Starship HLS	100 t-class reusable	$\sim$50 000	Projected
Fully reusable	FH-class × 7× multiplier	$\sim$10 000	Aspirational

Caveat: The "Starship" and "Fully reusable" scenarios assume successful development of fully or partially reusable lunar landing systems. SpaceX's current LEO cost target for Starship is \$78–94/kg with partial reusability, with a long-term target of \$10/kg; lunar surface costs include the substantial delta-v penalty ($\sim$6 km/s from LEO) and propellant depot operations that remain undemonstrated. These costs should be treated as bounding cases rather than planning baselines.

6.4 ISRU Extraction Cost Estimates

Extraction cost estimates derive from NASA technology development studies. Table 6 compiles published values with their underlying assumptions. Jones [24] estimates lunar oxygen delivered in a qualified tank at \$18 700/kg under the Falcon Heavy architecture; however, this assumes high-volume bulk delivery. At CLPS-era launch costs, the effective cost of oxygen in orbit far exceeds this figure.

Table 6: ISRU metal extraction cost estimates. O_2 values from Sanders [2] and Jones [24]; metal costs scaled from oxygen extraction as described in text.

Product	$C_{\mathbf{op}}$ (\$/kg)	Basis	Source
O_2 (from regolith)	500 000–2 000 000	Amortised, 10-yr	Sanders 2025
O_2 (optimistic)	∼18 700	FH-era bulk delivery	Jones 2023
Fe (H_2 reduction)	100 000–500 000*	Scaled from O_2	This work
Al–Si (MRE/FFC)	200 000–1 000 000*	Scaled from O_2	This work
Fe–Ti (ilmenite)	80 000–400 000*	High-Ti mare	This work

*Extrapolated from oxygen extraction costs; treat as order-of-magnitude.

Iron extraction from high-Ti mare ilmenite is projected to have the lowest cost due to the relatively simple hydrogen reduction chemistry and high ilmenite concentrations in mare regolith [1, 11]. The "This work" estimates scale O_2 extraction costs by the additional energy and processing complexity of metal reduction: Fe from ilmenite requires only H_2 reduction at ∼1 000°C (modest uplift), while Al–Si via MRE or FFC-Cambridge requires higher-temperature electrolysis (∼1 600°C) and more complex cell design (2–5× uplift). These are deliberately wide ranges reflecting order-of-magnitude uncertainty.

6.5 Break-Even Analysis

Figure 11 presents contour plots of the break-even operating cost (Eq. 8) across the launch cost–production volume parameter space for two IMLR scenarios: conservative (IMLR = 2, corresponding to ∼50% ISRU content) and optimistic (IMLR = 5, corresponding to ∼80% ISRU content).

Key observations:

1. At current demonstrated launch costs ($C_L \approx$ \$1M/kg), ISRU-integrated manufacturing breaks even at moderate production volumes ($N > 100$ components for IMLR = 2; $N > 63$ for IMLR = 5). At $N = 1{,}000$, the maximum viable operating cost reaches ∼900 k\$/kg, comparable to the upper range of oxygen extraction estimates in Table 6.

2. Counter-intuitively, the Starship-era scenario ($C_L \approx$ \$50k/kg) makes ISRU *harder* to justify economically: the capital cost of the metallurgical station (\$500M) now dominates, requiring $N > 2{,}000$ components (IMLR $= 2$) or $N > 1{,}250$ (IMLR $= 5$) before the operating cost budget turns positive. Cheap launch undercuts the economic case for in-situ production unless capital costs fall proportionally.

3. Higher IMLR values (more ISRU content) improve the capital amortisation rate (larger M_{ISRU} per component) but require validated alloy compositions and process windows; precisely the gap that SPARK calibration addresses.

4. The capital cost term dominates at low production volumes and low launch costs alike, emphasising the importance of designing modular, multi-purpose metallurgical stations rather than single-product facilities, and of reducing station mass through technology maturation.

6.6 Economic Amortisation via Dual-Purpose Mass

The break-even analysis above assumes that the Earth-shipped refractory fraction (M_{Earth}) is transported as dead-weight cargo: raw powder with no function other than serving as alloy feedstock. This is the worst case economically: every kilogram of refractory incurs the full launch penalty C_L. We now introduce a **dual-purpose mass** concept that eliminates this penalty entirely.

The gravity-margin dividend. Earth-launched spacecraft structures are sized for launch loads: quasi-static accelerations of 3–6g axial plus severe acoustic and random vibration environments during ascent through maximum dynamic pressure (Max-Q) [25]. Once a lander reaches the lunar surface, the gravitational load drops to 0.16g—roughly 20–40$\times$ below the design loads. Landing struts, structural frames, thermal shields, and mounting brackets transition from load-bearing members to dead weight. The structural mass fraction of a lunar lander is typically 15–25% of landed mass, representing tens to hundreds of kilograms of high-performance metal that serve no further purpose after touchdown.

Dual-purpose mass concept. Instead of shipping raw refractory powder as dead-weight cargo, we propose manufacturing critical spacecraft structural components (landing struts, mounting brackets, thrust-structure frames) from concentrated RHEA compositions, drawing on the broad compositional design space mapped by Gorsse et al. [26]. These components fulfil their primary mission duty (surviving launch loads, providing structural integrity during transit and landing) and are then **cannibalized** after landing to serve as a master-alloy seed for ISRU-blended manufacturing. The refractory elements (Mo, Nb, Ta, W) arrive on the lunar surface not as inert cargo but as functional hardware whose transport cost is entirely amortised by its primary mission role.

Master alloy dilution. The cannibalized RHEA components are not recast as pure refractory parts. Following the classical master-alloy paradigm used in grain refinement of light alloys [27], they are diluted with locally extracted bulk metals (Fe–Ti from mare ilmenite reduction or Al–Si from FFC-Cambridge processing) per the blending concept of Section 5.3. A single 100 kg RHEA landing strut, for example, acts as a master alloy seed for 400 kg of ISRU iron, yielding 500 kg of a medium-strength hybrid alloy (IMLR $= 5$) suitable for structural applications below $\sim$800°C. The dilution must remain within the single-phase BCC stability window to avoid embrittlement from Laves-phase formation (Fe_2Nb, Fe_2Ta); Figure 12 maps the phase evolution schematically. The target dilution window (75–90% ISRU by mass) keeps the hybrid composition within the BCC field while providing sufficient refractory content for solid-solution strengthening.

Modified cost model. Because the transport cost of M_{Earth} is entirely amortised by its primary mission role (delivering the payload to the lunar surface), its effective launch cost for the metallurgical process drops to zero. The ISRU manufacturing cost (Eq. 7) reduces to:

$$C_{\text{total}}^{\text{dual}} = C_{\text{cap}} + C_{\text{op}} \cdot M_{\text{ISRU}} \cdot N. \tag{9}$$

Setting this equal to the pure Earth-supply baseline ($C_{\text{total}}^{\text{Earth}} = C_L \cdot M_{\text{comp}} \cdot N$) and solving for C_{op}:

$$C_{\text{op}} < C_L \cdot \frac{\text{IMLR}}{\text{IMLR} - 1} - \frac{C_{\text{cap}}}{N \cdot M_{\text{ISRU}}}. \tag{10}$$

The critical difference from the standard break-even (Eq. 8) is the multiplier on C_L: the standard model has a coefficient of 1, while the dual-purpose model has $\text{IMLR}/(\text{IMLR} - 1)$, which is strictly greater than 1 for all $\text{IMLR} > 1$. For IMLR $= 2$ the multiplier is 2.0; for IMLR $= 5$ it is 1.25.

Economic consequences. Figure 13 compares the viable parameter space under both models. The dual-purpose approach produces three quantifiable improvements:

1. **Halved break-even volumes at current launch costs:** At $C_L \approx$ \$1M/kg (CLPS era), dual-purpose mass breaks even at $N > 50$ components (IMLR $= 2$), compared to $N > 100$ for dead-weight cargo. At $N = 1{,}000$, the maximum viable operating cost doubles from $\sim$900 k\$/kg to $\sim$1,900 k\$/kg, well within the range of current ISRU extraction cost estimates (Table 6).
2. **Starship-era viability restored:** In the standard model, Starship-era launch costs ($C_L \approx$ \$50k/kg) made ISRU uneconomical except at very high volumes ($N > 2{,}000$). The dual-purpose model halves this to $N > 1{,}000$ and, critically, the operating cost budget at $N = 1{,}000$ rises from -50 k\$/kg (unviable) to $\sim$0 k\$/kg (break-even), neutralising the "cheap launch kills ISRU" paradox identified in Section 6.5.

3. **Higher IMLR amplifies the advantage:** At IMLR $= 5$, the dual-purpose multiplier of 1.25 compounds with the larger ISRU mass fraction: at CLPS-era costs and $N = 1{,}000$, the viable operating budget rises from 938 k$/kg to 1,188 k$/kg.

Metallurgical validation requirements. Three research campaigns are needed to validate the dual-purpose mass concept:

1. **Dilution trajectory modelling:** CALPHAD calculations (Thermo-Calc with TCHEA or equivalent databases) must map the pseudo-binary phase equilibria between candidate RHEA compositions and ISRU bulk metals (Fe–Ti, Al–Si) to identify safe dilution windows and embrittlement cliffs where detrimental intermetallic phases (Laves, σ) dominate.
2. **Impurity tolerance assessment:** Lunar-extracted metals will contain unreduced silicates and dissolved oxygen. The master alloy must tolerate high O/Si pickup during the dilution melt without catastrophic embrittlement. Vacuum induction melting of Ni-superalloy revert material (up to 70% recycled content) has demonstrated that inclusion evolution can be managed through controlled atmosphere processing [28]; analogous protocols must be developed for RHEA-plus-ISRU melts.
3. **Process path definition:** Converting a finished structural component into powder feedstock for HP/LPBF/DED requires either localised gas atomisation or mechanical milling. The latter is compatible with the MA capability already present in the SPARK process chain (40 h milling demonstrated). Adding a crush/mill step to the lunar station architecture is straightforward given that MA is a baseline SPARK process.

Limitations. The dual-purpose concept assumes that (i) spacecraft structural components can be designed for dual service without compromising launch-load performance, (ii) the cannibalized alloy retains sufficient quality after service exposure (thermal cycling, micrometeorite impacts, surface contamination), and (iii) the reprocessing infrastructure (milling/atomisation, melting, blending) is available at the lunar station. These assumptions are classified as **Roadmap hypothesis** (Tier R) per Table 8. However, even a partial realisation, cannibalizing only a fraction of the landed structural mass, would substantially improve the ISRU business case.

6.7 Programmatic Implications

The economic analysis supports a phased approach to lunar metallurgy:

Phase 1 (Near-term, TRL 4–6): Earth-side process calibration (SPARK) establishes the parameter space for ISRU-compatible alloys. No lunar infrastructure required; cost is absorbed in existing technology development programmes.

Phase 2 (Mid-term, TRL 6–7): Demonstration-scale ISRU metal extraction ($\sim$10 kg/year) validates feedstock quality assumptions. Components remain Earth-supplied; ISRU metals used for non-critical applications (ballast, radiation shielding).

Phase 3 (Long-term, TRL 7–9): Production-scale metallurgical station ($\sim$100–1000 kg/year) enabled by reduced launch costs. Break-even achieved for structural components; hot-section parts may still require Earth-supply of pure refractory feedstock.

The terrane-aware framework provides the site-selection logic for Phases 2–3: mare sites for Fe–Ti feedstock (highest I_{FeTi}), highlands sites for Al–Si feedstock (highest I_{AlCa}), with the metallurgical potential indices guiding infrastructure placement to maximise IMLR for the target component portfolio.

6.8 Limitations of Economic Projections

All cost projections in this section carry substantial uncertainty and should be interpreted as bounding analyses rather than point estimates. Launch costs are subject to market dynamics and technology development outcomes; extraction costs are extrapolated from oxygen production studies and involve additional processing steps not fully characterised at scale; the model assumes fungible production volume whereas in practice component diversity may limit economies of scale; and discount rates, inflation, and opportunity costs are not modelled. The parametric framework nonetheless provides mission architects with a tool to evaluate ISRU-integrated manufacturing under their own cost assumptions and identifies the threshold conditions that would make terrane-specific metallurgy economically compelling.

7 TRL Roadmap and Qualification Gates

The roadmap below aligns the terrane-aware metallurgy framework with the SPARK Earth-side qualification timeline and broader ESA/international exploration milestones.

Stage 1: Earth-side calibration (TRL 3–5, current)

- SPARK screening campaign completed (February 2026): eleven compositions modelled via Thermo-Calc CALPHAD, nine experimentally produced via MA + HP, with XRD/EDS phase analysis and mini-tensile/hardness characterisation. Two candidates down-selected: SPARK-R1 (five-element BCC RHEA, anomalous T-strengthening) and SPARK-S1 (proprietary Ni-based superalloy, 1 250 MPa RT UTS). Microstructural characterisation (EBSD, TEM) of down-selected candidates is in progress; results expected by April 2026.

- Lock feedstock acceptance windows (particle size distribution, purity thresholds, oxygen content) that define the boundary conditions for future ISRU-derived powder substitution.
- Develop process–property relationships (DoE-based) from SPARK experimental data to establish predictive capability for density, defects, and microstructure across the HP→LPBF/DED chain.

Gate criterion: At least one RHEA system demonstrates strength retention at >1 000 K: *partially met*. SPARK-R1 retains 532 MPa at 1 000 K with anomalous strengthening. ERBURIK oxidising gas-flow exposure at ArianeGroup is planned for the next campaign.

Stage 2: Proxy feedstock and hot-fire validation (TRL 5–6)

- Produce proxy lunar feedstocks from terrestrial regolith simulants (e.g. JSC-1A for mare, NU-LHT for highlands) via laboratory-scale extraction and refining.
- Demonstrate that the SPARK-calibrated process chain (HP screening → LPBF/DED manufacture) produces acceptable microstructures when input powder composition is shifted to simulant-derived chemistry.
- Fabricate graded or blended components across at least two terrane-informed chemistry classes (Fe–Ti-based and Al–Ca-based) via LPBF and/or DED.
- Conduct hot-fire validation of RHEA throat inlay in BK-I sub-scale pre-combustion chamber at DLR P8 test bench (Lampoldshausen), targeting chamber pressures up to 380 bar at mixture ratios of 50–100 (planned for FIRST! follow-on funding).

Gate criterion: Process-window transfer demonstrated on at least two simulant-derived feedstocks with acceptable deviation in key mechanical properties from terrestrial-powder baselines; throat inlay survives oxidizing gas-flow test campaign.

Stage 3: Lunar architecture integration (TRL 6+)

- Define modular metallurgical station architecture compatible with Artemis and E3P surface infrastructure concepts, including feedstock beneficiation, powder preparation, HP/LPBF/DED processing, and post-processing modules.
- Integrate process–property models with in-situ feedstock characterization sensors for closed-loop process control adaptable to terrane-specific feed variability.

- Demonstrate reduced Earth-import dependence by quantifying the mass fraction of structural and functional components producible from each terrane class.

Gate criterion: End-to-end demonstration (simulant extraction → powder preparation → HP screening → LPBF/DED manufacture → component testing) in a representative thermal-vacuum environment.

8 Limitations

Current limitations include:

- **LP-GRS spatial resolution:** At 2° (∼60 km), each pixel integrates multiple lithologies. Finer-resolution data (e.g. from future neutron/gamma spectrometers or sample-return missions) would sharpen terrane boundaries.
- **ISRU feed quality:** Actual regolith processing will introduce beneficiation losses, grain-size fractionation, and contamination not captured by bulk orbital chemistry.
- **Domain shift:** Earth-side calibration of process windows assumes that ISRU-derived powders will have comparable morphology, flowability, and purity to terrestrial equivalents: an assumption that requires in-situ validation.
- **Mini-tensile geometry:** All SPARK mechanical data were obtained from mini-tensile specimens (∼0.5×1.2 mm cross-section) cut from HP compacts. These geometries are subject to surface-finish and edge effects that can suppress measured ductility and shift apparent UTS relative to standard ASTM specimens. The Inconel benchmark tested under identical geometry shows ∼738 MPa at RT, compared to the handbook value of ∼1 034 MPa, suggesting a systematic geometry correction of ∼0.7× may apply. Comparative rankings within the SPARK dataset remain valid because all specimens share the same geometry.
- **Index weights:** The current weights in Eqs. 2–4 reflect metallurgical domain knowledge rather than formal optimisation. The Monte Carlo sensitivity analysis (Section 4.3, $n = 10{,}000$) shows 100% ranking stability under ± 20% weight perturbation, but formal weight optimisation against ground-truth extraction yields,when available from demonstration missions,could narrow the confidence intervals.
- **Mixed CLR + z-score space:** Combining CLR-transformed oxides with z-scored traces in PCA creates a hybrid metric space in which variance contributions of the two groups are not directly comparable. An ILR-based approach [14] would avoid the singular covariance matrix inherent to CLR; CLR was retained here for interpretability.

- **Zero replacement:** Terrane-wise median imputation for below-detection oxides (2 859 zeros, predominantly highlands TiO_2) compresses within-terrane variance. Multiplicative replacement at a fraction of the detection limit is the more standard approach in the compositional data analysis literature and would better preserve distributional shape.
- **Composition non-disclosure:** Exact SPARK alloy formulations are withheld under the consortium IP agreement. This limits the ability of external researchers to reproduce specific compositions or compare directly with published results for named alloys. The alloy-family metadata (element count, phase identification, density) and full process–property data are provided to enable independent assessment of the framework logic.

9 Conclusion

A terrane-aware metallurgy framework was presented to map lunar geochemistry into manufacturability-access space and assign alloy-family/process strategies under uncertainty-aware qualification logic. Using 11 306 spatial pixels from the LP-GRS 2-degree global dataset, PCA on CLR-transformed compositions recovers the three fundamental geochemical gradients of the lunar crust (highlands vs. mare enrichment, mafic differentiation, and Ti variability) with 83.6% cumulative variance explained and stable bootstrap loadings ($\sigma \leq 0.007$). Three metallurgical potential indices translate these geochemical gradients into site-selection metrics for Fe–Ti, Al–Ca, and KREEP-bearing resources.

The SPARK experimental campaign screened nine compositions via MA + HP at 1 500°C, identifying SPARK-R1 (anomalous strength increase at 1 000 K, single-phase BCC) and SPARK-S1 (1 250 MPa RT UTS) as candidates for combustion-chamber applications, while demonstrating that eight-element refractory HESAs are too brittle for structural use in their current form. The brittleness of monolithic refractory HESAs motivates the ISRU blending concept: ISRU-derived ductile phases (Fe–Ti, Al–Si) could address the toughness deficit while reducing Earth-import mass via IMLR values of 5–10.

The strongest contribution is a defensible transfer methodology and TRL pathway rather than a claim of fully implemented lunar industrial capability.

Data and Code Availability

The LP-GRS 2-degree elemental abundance dataset is publicly available from NASA's Planetary Data System (PDS) Geosciences Node.[2] The complete analysis pipeline (preprocessing, PCA, index computation, Monte Carlo sensitivity analysis, phase diagram calculations, and all figure-generation scripts) is available at https://github.com/Darth-Hidious/wams2026-lunar-metallurgy un-

[2] https://pds-geosciences.wustl.edu/missions/lunarp/

der an MIT open-source license, together with the LP-GRS input data, COST507 thermochemical database, and the Jupyter notebook used to generate all figures and tables in this paper.

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Author Contributions (CRediT-style)

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Siddhartha Yash Kovid	Conceptualization, methodology, software, formal analysis, writing – original draft, project administration, supervision.
Kevin Grüning	Methodology, validation, investigation, writing – review & editing.
[Add author]	*[Add contribution terms]*

Conflict of Interest

The authors declare no conflict of interest unless otherwise stated by institutional policy.

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Table 8: Evidence classification of key assertions. Each substantive claim is mapped to one of four tiers: **Measured** (M) = published experimental data; **Model-derived** (D) = computed from measured data via standard methods; **Assumed** (A) = expert judgement requiring future validation; **Roadmap** (R) = no experimental demonstration yet.

#	Assertion	Tier	Source / Basis
1	LP-GRS oxide and trace-element concentrations at 2° resolution	M	Prettyman et al. 2006
2	Three operational resource classes (Highlands / Mare / KREEP)	A	Paper's thresholds, inspired by Jolliff 2000
3	CLR + PCA recovers 83.6% variance in three PCs	D	Standard CoDa + PCA on LP-GRS data
4	PC loadings correspond to established geochemical gradients	D	Validated against lunar petrology literature
5	Silhouette = 0.300, DBI = 1.247 for three classes	D	Computed from 3-PC score space
6	Metallurgical index weights (I_{FeTi}, I_{AlCa}, I_{KREEP})	A	Expert judgement; MC sensitivity (Sec. 4.3) shows 100% ranking stability
7	H_2 reduction of ilmenite → Fe + O_2	M	Sanders 2013; Anand 2012
8	Molten regolith electrolysis → Fe–Si alloy + O_2	M	Sanders 2025; Lomax 2020
9	FFC-Cambridge process → Al–Si structural alloys	M	Lomax 2020
10	SPARK RHEA compositions (SPARK-R and SPARK-H series)	M	SPARK PR2, this work
11	HP process windows for SPARK compositions	M	SPARK PR2 + published literature
12	RHEA YS retention above 700°C vs. Ni-superalloys	M	Senkov 2018; Senkov & Miracle 2018
13	ISRU-extractable mass fraction of candidate alloys	D	Atomic-weight calculation on known compositions
14	Blending ISRU bulk metals with Earth-shipped refractories	R	No experimental demonstration
15	Process-window transfer to ISRU-derived powders	R	Extrapolated from SPARK calibration
16	Modular lunar metallurgical station	R	Architecture concept only
17	SPARK-R1 anomalous strengthening at 1 000 K	M	SPARK mini-tensile data (this work)
18	HESA brittleness and down-selection	M	SPARK processing observations (this work)
19	XRD phase confirmation (BCC for SPARK-R1, FCC+MC for SPARK-S1)	M	SPARK XRD (this work)
20	Mini-tensile geometry correction ($\sim 0.7\times$ vs ASTM standard)	D	Comparison with Inconel handbook value

Table 9: PCA loadings for the first three principal components (bootstrap mean $\pm\ 1\,\sigma$ from 200 sign-aligned resamples). Dominant loadings ($|w| > 0.30$) are set in bold.

	PC1 (53.1%)	**PC2** (17.4%)	**PC3** (13.2%)
FeO	0.31 ± 0.006	-0.35 ± 0.018	-0.39 ± 0.019
TiO_2	0.37 ± 0.006	-0.11 ± 0.028	0.61 ± 0.006
Al_2O_3	-0.39 ± 0.005	-0.01 ± 0.010	0.07 ± 0.016
CaO	-0.35 ± 0.006	-0.39 ± 0.016	-0.24 ± 0.023
MgO	0.03 ± 0.007	0.74 ± 0.021	-0.44 ± 0.034
SiO_2	-0.40 ± 0.004	-0.31 ± 0.012	-0.22 ± 0.016
K	0.40 ± 0.005	-0.22 ± 0.014	-0.28 ± 0.015
Th	0.42 ± 0.005	-0.12 ± 0.014	-0.29 ± 0.011

Table 10: Terrane-to-alloy/process compatibility matrix. Evidence classes follow Table 8. See footnotes for critical caveats.

Terrane	**Primary metals**	**Candidate alloy families**	**Extraction / mfg. route**	**Key risks**	**Evidence class**
Mare (Fe–Ti)	Fe, Ti, O_2	RHEA (Fe–Ti base + Earth-supplied Mo/Nb)[a]; structural steel	H_2 reduction of ilmenite; MRE $\rightarrow$ HP screening $\rightarrow$ LPBF/DED	Feed purity and morphology; ilmenite separation efficiency	M (extraction) + R (alloy blending)
Highlands (Al–Ca)	Al, Si, (Ca)[b]	HEA (Al-based + Earth-supplied Cr/Co); Al–Si structural alloys	FFC-Cambridge; MRE $\rightarrow$ HP	Oxide reduction yield; impurity tolerance in Al–Si system	M (extraction) + R (alloy)
KREEP-bearing	Fe, Ti + K, Th, U, REE	Functional RHEA layers; radiation shielding[c]	Combined Fe–Ti extraction + selective recovery of incompatibles	Radioactive material handling; very low REE concentration in regolith	A (extraction) + R (applications)

[a] Lunar ISRU produces bulk Fe and Ti; refractory elements (Mo, Nb, Ta, W) must be Earth-supplied until extraction routes for these elements are developed.
[b] Calcium is removed during electrolysis (remains in slag/electrolyte); usable outputs are Al–Si and Fe–Si multiphase alloys [22].
[c] Radiation-shielding potential derives specifically from Th and U (high-Z, effective gamma absorbers), not from the broader KREEP package (K, REE, P are low-to-moderate-Z).

Table 11: Hot-pressing and sintering process windows for SPARK compositions and representative published RHEA systems. MA = mechanical alloying; AM = arc-melted ingot crushed to powder; HP = vacuum hot pressing. SPARK compositions are identified by programme designators; exact formulations are withheld under the consortium IP agreement.

Alloy / Designation	T **(°C)**	P **(MPa)**	**Hold**	**Route**	**Density (g/cm^3)**	**Source**
Published literature baselines						
MoNbTaW	1 800	80	3 min	AM + SPS	>99% TD	Senkov 2011
MoNbTaVW	1 600	60	5 min	MA + SPS	97–99% TD	Senkov 2018
HfNbTaTiZr	1 300	50	20 min	MA + SPS	98% TD	Yao 2018
MoNbTaTiV	1 500	50	10 min	MA + SPS	>99% TD	Cao 2019
$AlMo_{0.5}NbTa_{0.5}TiZr$	1 400*	50*	10 min*	AM	—	Senkov 2018
SPARK systems (this work, MA + HP at Bimo Tech)						
SPARK-R1 ★	1 500	30	2 h	MA + HP	11.45	This work
SPARK-R2	1 500	30	2 h	MA + HP	11.68	This work
SPARK-R3	1 500	30	2 h	MA + HP	12.84	This work
SPARK-R4	1 500	30	2 h	MA + HP	11.32	This work
SPARK-R5	1 500	30	2 h	MA + HP	12.19	This work
SPARK-R6	1 500	30	2 h	MA + HP	12.92	This work
SPARK-H1	1 500	30	2 h	MA + HP	10.24	This work
SPARK-H2	1 500	30	2 h	MA + HP	9.39	This work
SPARK-H3	1 500	30	2 h	MA + HP	9.72	This work
SPARK-S1 ★	1 260	30	2 h	MA + HP	—	This work

*Originally reported as arc-melted; SPS parameters estimated from similar compositions.

★ Down-selected candidate.

Table 12: Tensile strength (UTS, mini-tensile geometry) at room temperature and 1 000 K for SPARK compositions and published RHEA baselines, with Vickers hardness and conservative ISRU-extractable mass fraction. SPARK compositions identified by programme designators; exact formulations withheld. ISRU set = {Fe, Ti, Al, Si, Mg}; for SPARK compositions, ISRU% is reported as an upper bound for the relevant alloy family.

	UTS (MPa)					
Alloy / Designation	**RT**	**1 000 K**	**HV2**	**ISRU (wt%)**	**Notes**	**Source**
Published baselines (compressive YS, standard specimens)						
MoNbTaW	1 058	548†	—	0		Senkov 2011
MoNbTaVW	1 246	659†	—	0		Senkov 2018
$AlMo_{0.5}NbTa_{0.5}TiZr$	2 000	745†	—	19		Senkov 2018
Inconel 718 (literature)	1 034	—	—	20	Tensile	Literature
SPARK systems (this work, mini-tensile UTS)						
SPARK-R1 ★	504	532	1 219	0	Anomalous $T\uparrow$	This work
SPARK-R2	368	—	1 170	0		This work
SPARK-R3	272	252	620	0		This work
SPARK-R4	–	—	330–470	≤10	No tensile data	This work
SPARK-R5	171	—	720–850	0	Brittle	This work
SPARK-R6	191	—	1 000–1 100	0	Brittle	This work
SPARK-H1	–	—	500–670	≤8	Brittle	This work
SPARK-H2	–	—	∼705	—	Too brittle to test	This work
SPARK-H3	165	—	840	—	Brittle	This work
SPARK-S1 ★	1 250	510	570	—	Proprietary	This work
Inconel (HP, same geometry)	738	974‡	470	∼20	HP benchmark	This work

★ Down-selected candidate.

† Literature values; compressive yield strength, standard specimens.

‡ Apparent strength increase at 1 000 K may reflect γ'' precipitation hardening and/or specimen geometry effects; see text.

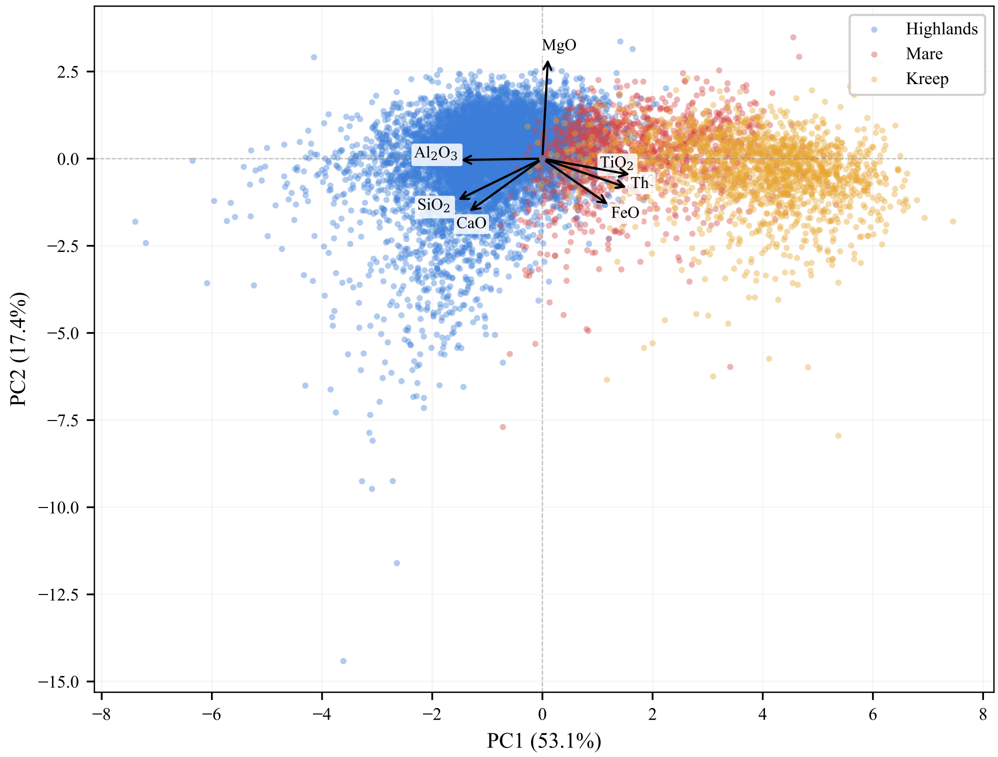

Figure 1: PC1–PC2 biplot of LP-GRS 2° pixels, coloured by operational terrane class. Highlands (blue) cluster at negative PC1; mare (red) and KREEP-bearing (orange) pixels spread along positive PC1 with overlap in the KREEP–mare transition zone.

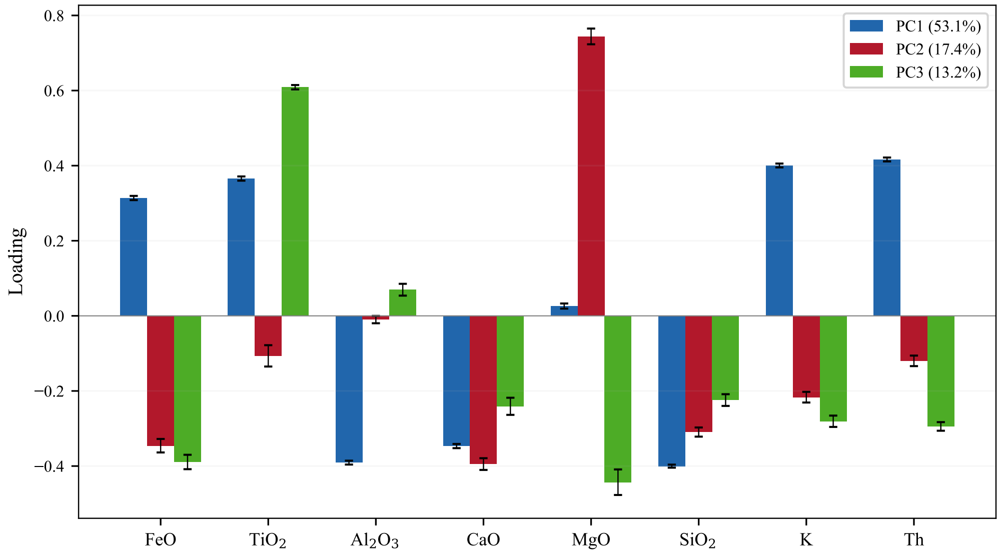

Figure 2: Loading vectors for PC1–PC3. PC1 separates plagioclase-associated elements (Al_2O_3, CaO, SiO_2 negative) from mare/KREEP indicators (FeO, TiO_2, K, Th positive). PC2 is dominated by MgO. PC3 distinguishes high-Ti from low-Ti mare basalts.

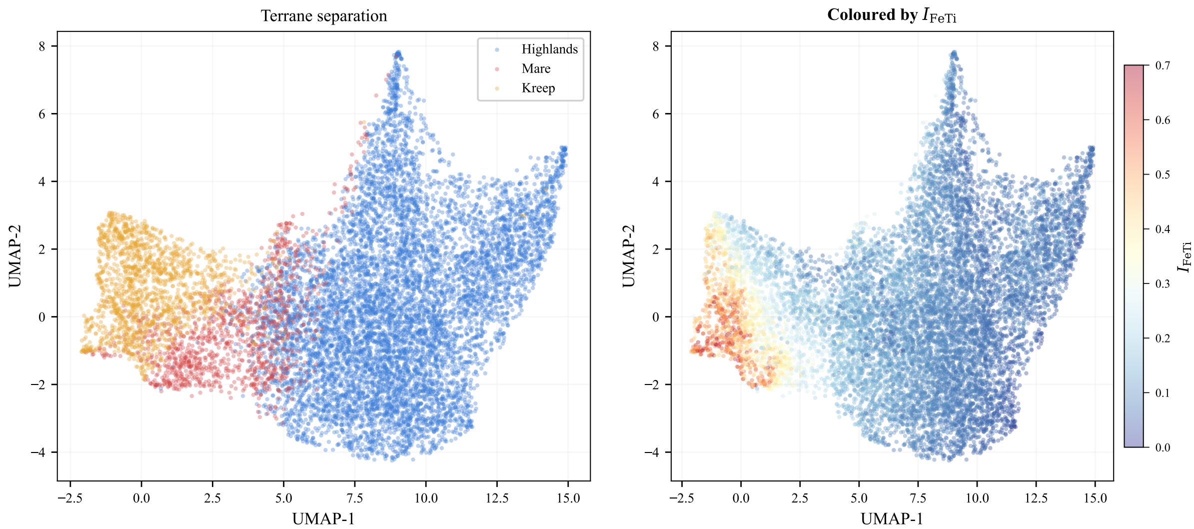

Figure 3: UMAP embedding of the 8-dimensional geochemical feature space ($n_{\text{neighbors}} = 30$, $d_{\text{min}} = 0.3$). **Left:** coloured by terrane class. The terranes form a continuous manifold rather than discrete clusters, with highlands (blue) as the dominant mass, mare (red) as a distinct arm, and KREEP (orange) bridging the two. **Right:** coloured by I_{FeTi}. The metallurgical index gradient maps smoothly onto the manifold, confirming that it captures genuine geochemical variation. Separability metrics (silhouette = 0.291, DBI = 1.089) are consistent with PCA (0.300, 1.247).

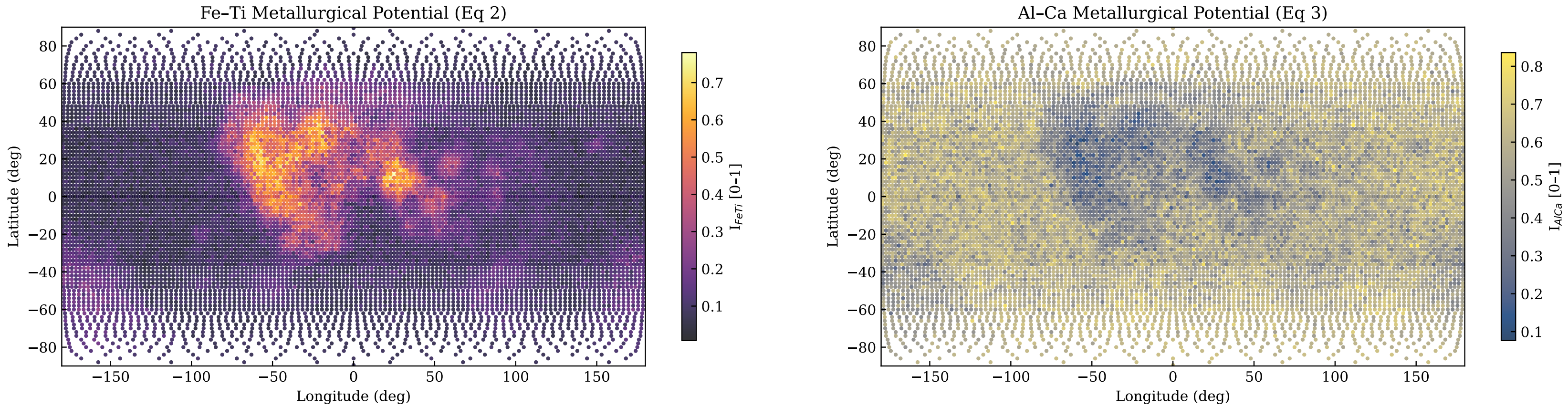

Figure 4: Selenographic projections of the Fe–Ti metallurgical potential index (**left**, I_{FeTi}) and the Al–Ca index (**right**, I_{AlCa}). High I_{FeTi} concentrates in the major nearside mare basins (Oceanus Procellarum, Mare Imbrium, Mare Tranquillitatis); high I_{AlCa} dominates the far-side feldspathic highlands.

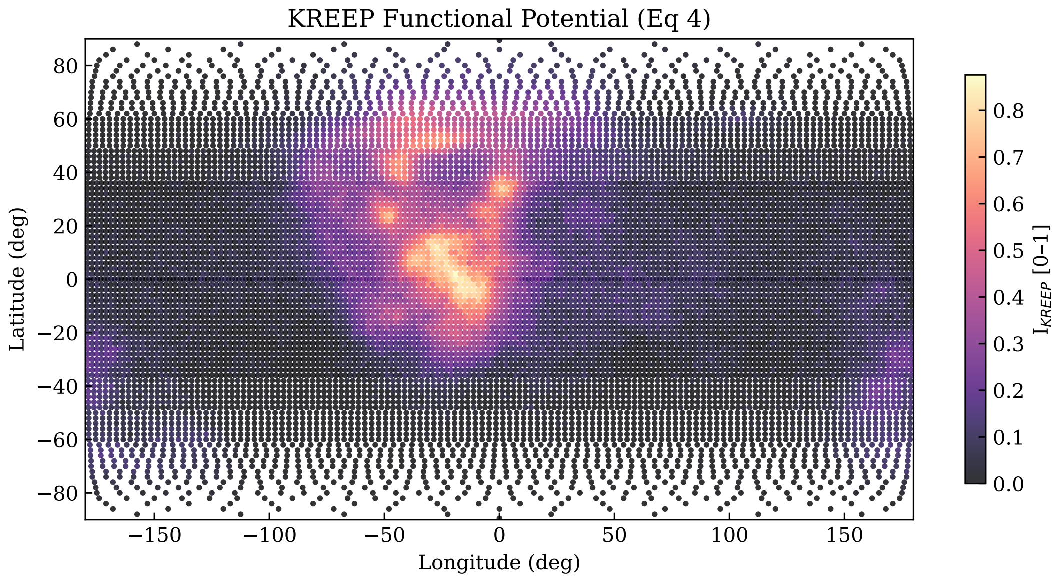

Figure 5: Selenographic projection of the KREEP metallurgical potential index (I_{KREEP}). The highest values concentrate in the Procellarum KREEP Terrane on the lunar nearside, co-located with elevated I_{FeTi} (cf. Figure 4).

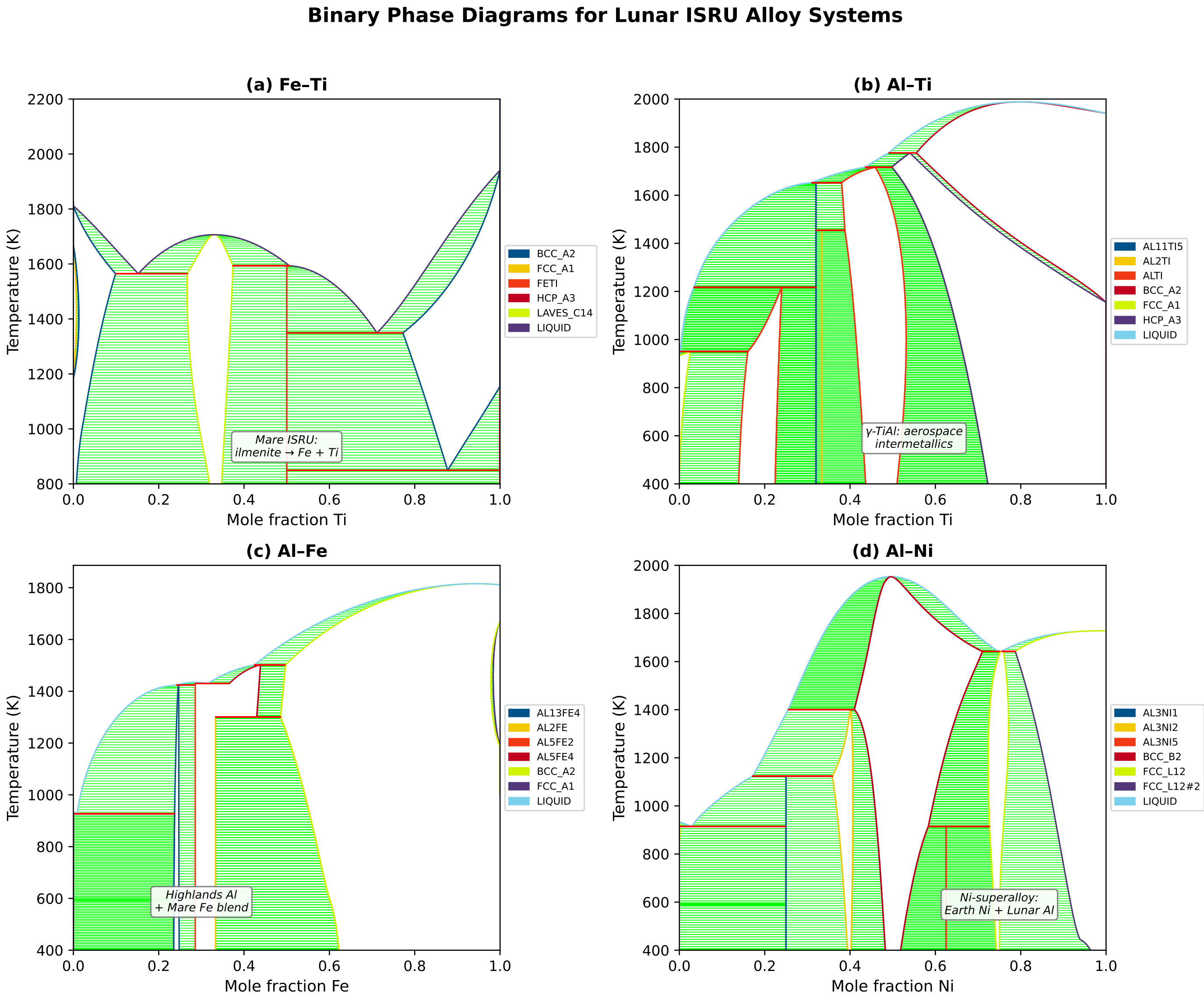

Figure 6: Binary phase diagrams for ISRU-relevant alloy systems, computed using the COST507 thermochemical database and pycalphad. (a) Fe–Ti: products of mare ilmenite reduction, showing the FeTi intermetallic and Laves C14 (Fe_2Ti) phase; (b) Al–Ti: TiAl intermetallics accessible from highlands Al + mare Ti, relevant to aerospace gamma-TiAl alloys; (c) Al–Fe: cross-terrane blend of highlands Al with mare Fe; (d) Al–Ni: Ni-superalloy route combining Earth-supplied Ni with lunar Al, showing the Ni_3Al (γ') formation region. These diagrams bound the compositional space for the graded architecture discussed in Section 5.3.

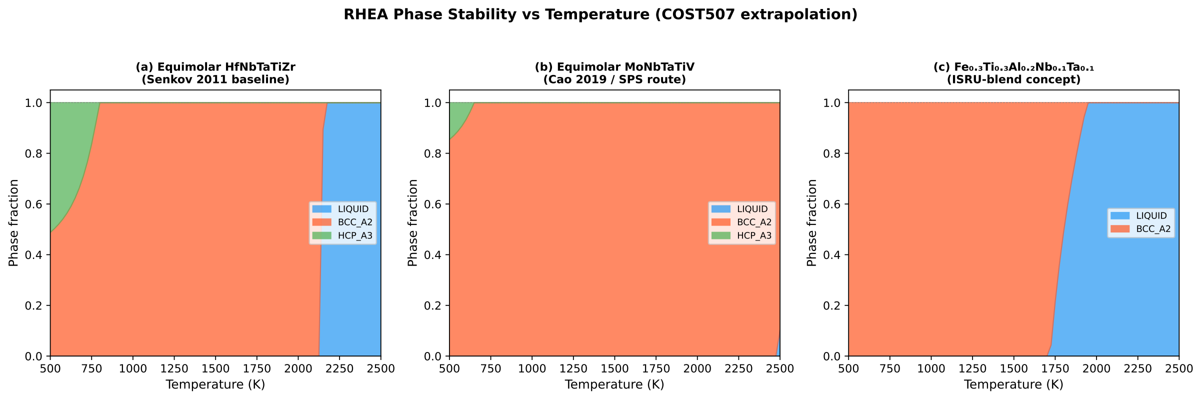

Figure 7: Phase fraction vs. temperature for candidate RHEA and ISRU-blend compositions, computed from COST507 binary interaction parameters extrapolated to multi-component space. (a) Equimolar HfNbTaTiZr (Senkov 2011 baseline): predicted single-phase BCC above ∼800 K; note that COST507 likely lacks Hf–Nb, Hf–Ta, and Nb–Ta interaction parameters, so ideal-solution mixing is assumed and single-phase BCC is a default rather than a validated prediction; experimental XRD confirms BCC phase stability for this composition [20]; (b) Equimolar MoNbTaTiV (Cao 2019 SPS route): predicted BCC across the full solid range, subject to the same database limitation; (c) ISRU-blend concept $Fe_{0.3}Ti_{0.3}Al_{0.2}Nb_{0.1}Ta_{0.1}$: BCC stable to ∼1 800 K but with lower melting point than pure refractory RHEAs, illustrating the melting-point depression trade-off discussed in Section 5.3. *Note:* these are COST507 extrapolations, not TCHEA-grade calculations; SPARK project data should be preferred where available.

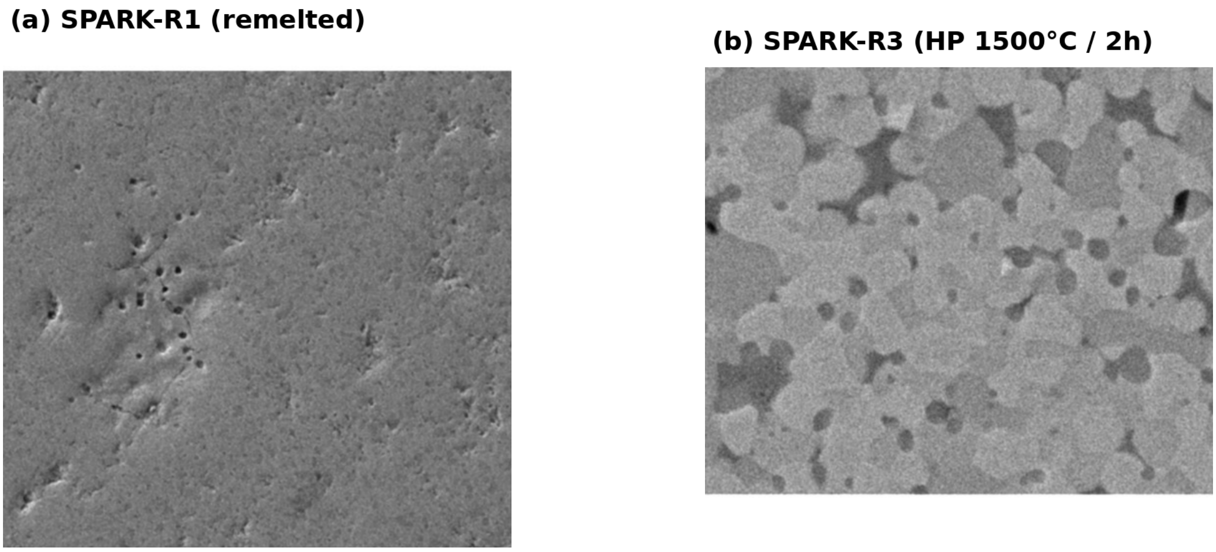

Figure 8: SEM backscatter micrographs of two SPARK five-element RHEAs. (a) SPARK-R1 (remelted): uniform contrast with minor residual porosity, consistent with the single-phase BCC structure confirmed by XRD. (b) SPARK-R3 (HP 1 500°C / 2 h): pronounced two-phase microstructure with bright matrix grains and darker precipitate regions. EDS mapping (not shown) confirmed segregation of one constituent to grain boundaries in SPARK-R3, whereas SPARK-R1 showed homogeneous distribution of all five elements. This microstructural contrast informed the down-selection of SPARK-R1 over SPARK-R3.

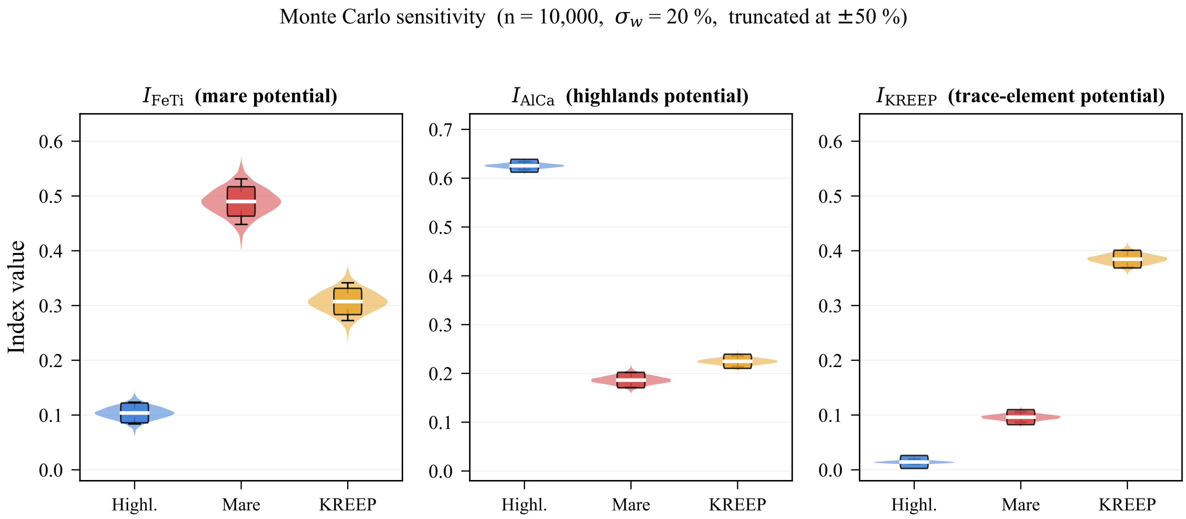

Figure 9: Monte Carlo sensitivity analysis of metallurgical potential indices. Violin plots show the distribution of terrane-mean index values across 10 000 weight perturbations ($\sigma = 0.20$, truncated at $\pm 50\%$). Coloured boxes denote IQR with white median line; whiskers span the 5th–95th percentile range. Terrane rankings are perfectly stable: highlands leads on I_{AlCa}, mare on I_{FeTi}, and KREEP-bearing on I_{KREEP} in 100% of realisations.

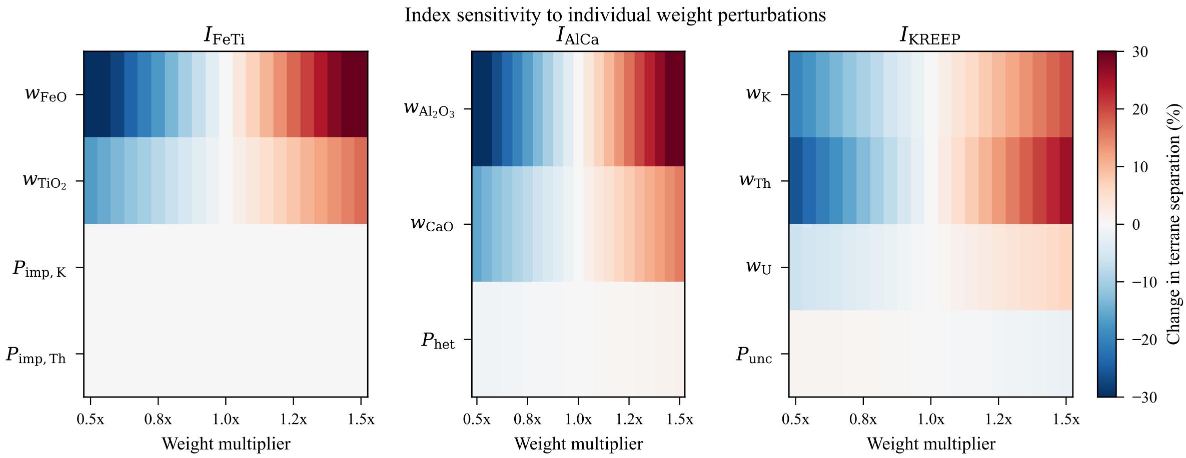

Figure 10: Sensitivity of inter-terrane separation to individual weight perturbations. Each cell shows the percentage change in the spread between the highest- and lowest-scoring terranes when a single weight is multiplied by the indicated factor ($0.5\times$ to $1.5\times$). Red: increased separation; blue: decreased separation. The dominant composition weights (w_{FeO}, $w_{\mathrm{Al_2O_3}}$, w_{K}) drive the largest effects ($\pm 25\%$), while penalty terms (P_{imp}, P_{het}, P_{unc}) have negligible influence.

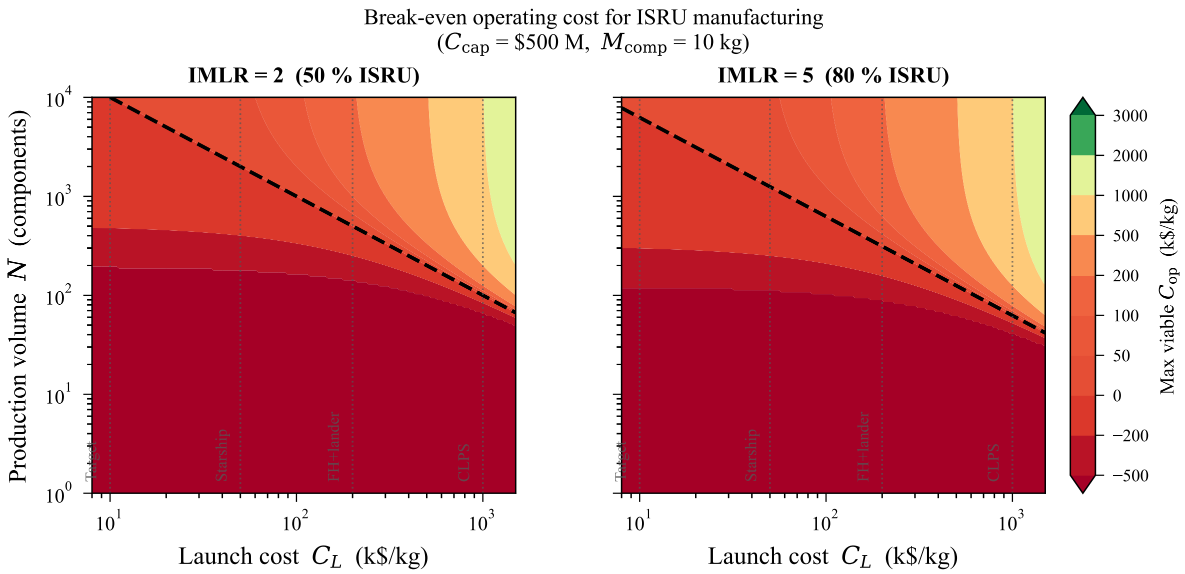

Figure 11: Break-even operating cost contours for ISRU-integrated manufacturing (dead-weight cargo model). **Left:** IMLR $= 2$ (50% ISRU). **Right:** IMLR $= 5$ (80% ISRU). Capital cost assumed \$500M (including launch of metallurgical station). Green regions indicate parameter combinations where ISRU-integrated manufacturing is cost-competitive with pure Earth-supply. The dashed black line marks the break-even boundary ($C_{\mathrm{op}} = 0$). Vertical dotted lines mark launch-cost scenarios from Table 5.

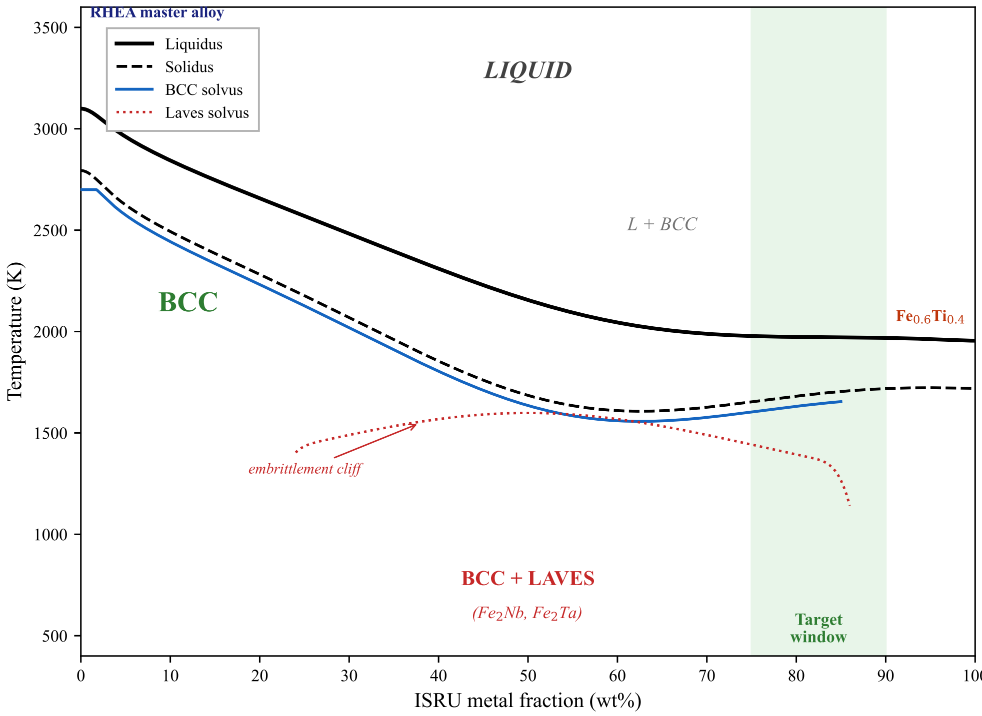

Figure 12: Schematic pseudo-binary phase diagram for the dilution trajectory from a concentrated RHEA master alloy (left) to ISRU-derived Fe–Ti bulk metal (right). Phase boundaries are constructed from the Fe–Ti binary (COST507, well-characterised) with refractory interaction effects informed by published HEA literature [7, 9]. The green band marks the target dilution window (75–90 wt% ISRU), where the hybrid alloy is expected to retain a single-phase BCC structure with solid-solution strengthening from dissolved refractory elements. The Laves-phase field (Fe_2Nb, Fe_2Ta) at intermediate compositions represents the embrittlement cliff that must be avoided. *Note:* this diagram is illustrative; quantitative phase boundaries require TCHEA-grade CALPHAD calculations for specific RHEA+Fe–Ti pseudo-binaries.

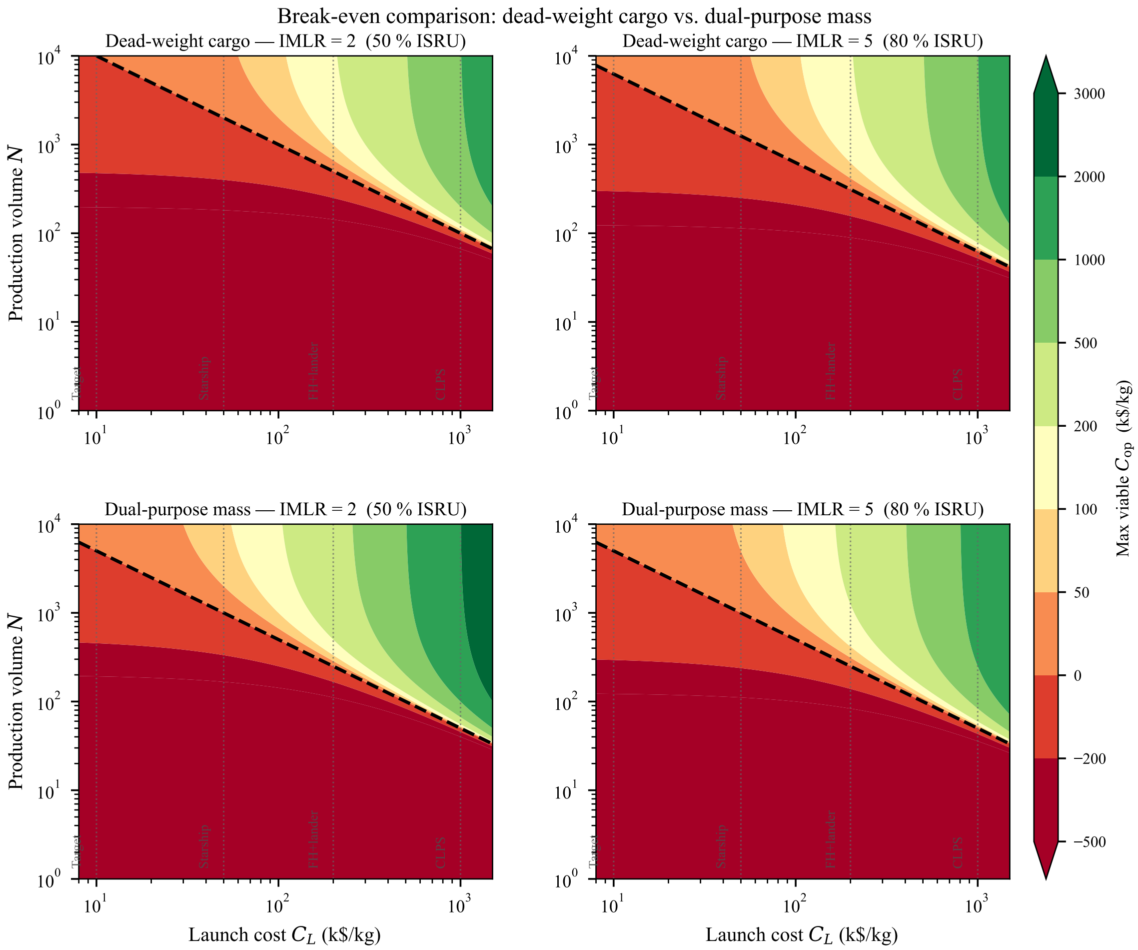

Figure 13: Break-even comparison: dead-weight cargo (top row) vs. dual-purpose mass (bottom row). **Left column:** IMLR $= 2$ (50% ISRU). **Right column:** IMLR $= 5$ (80% ISRU). In the dual-purpose model, the Earth-shipped refractory mass serves a primary structural function during transit and landing; its launch cost is fully amortised, eliminating the $C_L \cdot M_{\text{Earth}} \cdot N$ term from the cost model (Eq. 9). The resulting C_L multiplier of $\text{IMLR}/(\text{IMLR} - 1) > 1$ shifts the break-even boundary leftward and downward, expanding the viable green region into the low-launch-cost territory where the standard model predicts ISRU to be uneconomical.